A Satellite and Radar Analysis of the May 23. 2025 Tornado Event in Akron, CO

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ABSTRACT

On May 23, 2025, a long-lived supercell developed along the Colorado-Wyoming border and produced multiple tornadoes near Akron, Colorado, including an EF2 and EF1 that caused structural damage and disrupted local infrastructure. Despite relatively weak synoptic forcing, the storm evolved within a highly favorable mesoscale and terrain-influenced environment. This study examines the event using a multi factor approach that integrates satellite imagery, radar products, synoptic variables, and mesoscale variables to determine how local terrain features, boundary interactions, and storm scale processes contributed to tornadogenesis during this event. Satellite observations from GOES-16 water vapor, visible, and infrared bands showed consistent evidence of vertical development, overshooting tops, and upper-level divergence signatures that were aligned with the consistent updraft. Mesoscale variables visualize the importance of moisture transportation, baroclinic zones, and low-level shear, which were all enhanced by the influence of the Cheyenne Ridge. Radar reflectivity and velocity fields show a well-known supercell structure, with multiple boundary collisions and well-defined rear and forward flank downdrafts. Dual Pol products, such as correlation coefficient and spectrum width, visualize supporting evidence of strengthening rotation, increase in hydrometeor size, and shear values near the surface as the storm interacted with the outflow boundaries. Taken together, these products show that tornadogenesis, in this case, was heavily influenced by terrain features, boundary interactions, and small-scale processes happening near the surface. This analysis shows how an integration of satellite and radar products can be a helpful factor in understanding storm evolution in environments where synoptic scale forcing is lacking.

1. Introduction

On May 23, 2025, a supercell produced a series of tornadoes near Akron, Colorado, including an EF2 and an EF1 that snapped power poles, damaged trees, and damaged one farmstead. This event is worth researching due to its long lifespan, despite not having very sufficient values of forcing from a synoptic scale standpoint. It relied on more mesoscale and local features, along with local influences from terrain. This case study will be central to a satellite and radar analysis of the event, with the radar analysis following the midterm period. Currently, the analyzed satellite data gives a detailed view of how this storm grew and evolved in the unique environment it was in. One influence that the storm had was the Cheyenne Ridge, which is located in northwestern Colorado. The ridge helped shape and deform low-level winds, which created a zone of local convergence where lift could be sustained. In similar situations like these, where there is not sufficient synoptic support, terrain features such as the Cheyenne Ridge can play a huge role in storm initiation.

This analysis pays close attention to the evolution seen in the analyzed imagery, particularly in visible, infrared, and water vapor bands. Visible imagery helps to show where and when the storm began, as well as cloud features such as overshooting tops, which point to strong updrafts. Infrared imagery is useful in seeing cloud top temperatures, which align with how far the system grows in a vertical sense. Water vapor imagery helps to fill in where the moisture values are being transported and to identify features such as jet streaks that can have an influence on the storm. These bands of imagery allow for these factors to be analyzed with a closer eye and applied to the study of this storm. By analyzing these different bands of imagery together, a more complete picture of the Akron supercell can be formed. Each of these bands emphasizes different aspects of this storm and shows how it was able to intensify with time. The insights that were gained from this analysis will be able to contribute to a larger understanding of how satellite imagery can be utilized to identify the patterns that are present in severe storm development.

2. Data and Methodology

The data utilized for this analysis were primarily obtained from the MCFETCH satellite archive, along with archived satellite data from Iowa State University and College of DuPage. Data extraction was performed using Python in the Jupyter Notebook environment, with CLIMLAB downloaded into the environment. In addition, the NOAA Climate Toolkit was utilized to help visualize and interpret some of these satellite images to analyze them from a broader perspective of the storm. Radar analysis was completed using GR2 Analyst and GRLevel3 software, which allowed for data to be visualized as well as isosurface analysis. Data was downloaded utilizing the Amazon Cloud and was then interpreted into the mentioned software. This was not only used to determine storm motion and strength, but isosurface analysis also allowed for analysis of updraft tilt, mesocyclone size, and vertical continuity of rotation. For the synoptic and mesoscale analysis, data were sourced from archived surface analyses from the Weather Prediction Center (WPC) to help assess the conditions near the surface at the time of the event. To assess the mesoscale and synoptic parameters on a closer scale, the SPC Mesonalysis Archive was used to obtain maps showing some of these important parameters. Maps were captured the day of the event at 21z and 00z in order to analyze the environment as the storm formed. Archive forecast discussions, coming from the Iowa Environmental Mesonet, were reviewed to provide any additional insight. Additionally, surface observations from the College of DuPage (COD) were used to identify areas of pressure changes and wind shifts. A hand analysis of this surface analysis was completed on the morning of the event at 12Z. All together, these sources allowed for a concise and efficient analysis of this event, integrating multiple forcing mechanisms and factors that influenced the tornado.

3. Storm Analysis

3.1 Synoptic Analysis

On May 23, 2025, there was a large, upper-level ridge situated over the Rockies and into the Great Plains, with the axis extending southeastward into the Great Plains and Texas. Within this large ridging pattern, there was a prominent negatively tilted shortwave trough at 300mb, with the axis extending from central Wyoming down into northeastern Colorado. The 300mb level at 23Z showed this feature well, and the observed shortwave is most likely to be terrain influenced coming off the Rocky Mountains or the Denver Convergence Vorticity Zone (DCVZ) (Fig. 34). Along with the upper-level shortwave trough, there are larger localized regions of upper-level divergence, which are positioned downstream across northeastern Colorado (Fig. 21). These winds aloft are relatively weak, with the upper-level flow over Akron settling at around 50kt. The areas of localized divergence sat downstream of the shortwave, placing the bullseye of higher divergence values right over the northeastern corner of Colorado. This localized upper-level divergence is efficient at assisting in large-scale upward motion in the region. Along with the elevated values of local divergence and the shortwave, there are elevated values of relative humidity (RH), greater than or equal to 70 percent, which means that the atmosphere is holding about 70% of the maximum amount of water vapor it can sustain at that given temperature. These elevated values of RH allow there to be support for cloud development and sustained upward motion, but this is reliant on outside lifting mechanisms. Also noted at the 700mb level is a wind shift over the area from Southwesterly to Westerly, which can indicate the passage of the shortwave trough that was mentioned earlier (Fig. 21). Contrary to these values, at 700-400mb, there seems to be a high level of Negative Vorticity Advection (NVA), which is not ideal for a tornadic environment, as these negative values are associated with downward motion. These negative values are centered almost directly over Akron, CO.

Moving down to 850mb, there are gradients of height, temperature, and dewpoints, which all seem to be strengthening over the forecast area (Fig. 22). The tightening height gradient shows that there are strengthening winds in the lower levels, which can mean more concentrated values of lift along the gradient, which is a result of air being pressed against the Rockies. Similarly, the tightening temperature gradient can be a sign of a strengthening baroclinic zone, which in turn increases horizontal vorticity values in the same region. This temperature gradient would also be a signature of an increased area of upward vertical motion. In addition to the gradients, there is an area of Cold Air Advection (CAA) that overlaps with these strengthening gradients (Fig 24). This adds another layer to the temperature across the boundary, which further enhances the sharpness of the baroclinic zone. When combined with strengthening low-level winds, this created an environment that was favorable for horizontal vorticity to be stretched into the vertical by a developing updraft. Also near the surface, there is an observed area of lower pressure in southeastern/south-central Colorado. This low caused elevated values of Theta E to wrap around its circulation, which is indicative of strong transportation of moisture into the target area, even though mid and high-level forcing is unfavorable (Fig. 27).

3.2 Mesoscale Analysis

Beginning down near the surface on May 23, 2025, at 17-20z, there is an observed pressure gradient that is tightening over the northeastern corner of Colorado, with the pressure values ranging from 1004hPa at the lowest, and 1016hPa at the highest (Fig. 22). The tightening of this gradient showed that there was an enhancement of lower-level flow in this region, and it increased as the day moved forward. In southeastern to southcentral Colorado, there was a primary area of low pressure close to the surface, with a secondary low developed in eastcentral Wyoming. At 18Z-21Z, a dryline extended southward from the southeastern Colorado low into New Mexico and the Texas panhandle, and a warm front extending southward from the low in Wyoming into northeastern Colorado along its shared border with Nebraska and Kansas. These localized boundaries provided sources of lift and convergence in the event region. Staying close to the surface, values of surface moisture convergence were at $10 * 10^{-5}s - 1$ by 23z, which means that there are focused enhanced values of ascent in the area due to converging winds and moisture advection. This, in turn, forces rising motion and enhances the possibility of initiation. Part of this enhancement can be linked to the influence of the Cheyenne Ridge (Fig. 33), which has a local influence on flow in the lower levels. As the southwesterly winds hit the ridge, convergence was further focused into northeastern Colorado. At the same time, the lifted condensation levels (LCLs) were lowered, sitting around 1750m (Fig. 26). Low LCLs usually are a signal of a moist boundary layer, which allows lifted air parcels to rise and reach saturation faster.

Values of Storm Relative Helicity (SRH) from 0-3km started out weak, but they increased to around $200m^2 s^{-2}$ at 23Z (Fig. 28) while 0-6km shear sat around 50kt, which lines up with the time of the event, and these elevated values of shear indicate a favorable environment for rotating updrafts. The 0-3km Energy Helicity Index (EHI), which combines CAPE and SRH to look at the potential for rotating updrafts that will produce storms or tornadoes, was sitting around a value of 5, which indicates that the environment would be supportive of development due to the overlap of buoyancy and shear values. The Mixed Layer Convective Available Potential Energy (MLCAPE) values were coined around 2000 J/kg^{-1}, which signifies a moderately unstable environment that would be able to sustain strong updrafts necessary for storm development. Surface Based CAPE (SBCAPE) values climbed to values as high as 3000j/kg, which is a substantial amount and enough to have a hand in enhancing the development of these storms (Fig.23). Increased values of CAPE near the surface can contribute to the probability of horizontal vorticity being stretched into the vertical. In addition to these factors, there were multiple outflow boundaries (OFBs) present from the developing storm. A boundary moving southeastward with the developing storm intersected with a northward-moving boundary associated with storm development to the south. The interactions between these OFBs enhanced horizontal vorticity as well as localized vertical stretching, which was important in the intensification of the Akron supercell.

3.3 Topographically Influenced Storms

The Cheyenne Ridge, located along the CO-WY border, has been documented to modify low-level (LL) wind patterns and convergence near the surface across northeastern CO. The ridge redirects southeasterly flow patterns northward and upward, which promotes localized zones of lift, upward motion, and horizontal shear that can have a hand in enhancing convective initiation. Previous work by Bann, Weiland, and McAnelly (2003) and Wilczak and Glendening (1988) has shown that topographic features such as these can create mesoscale environments that are favorable for severe convection, even without the influence of strong synoptic scale forcing. In these environments, air goes up slope and is forced to converge with the downslope flow on the lee side of the ridge, which creates areas of baroclinic vorticity and enhanced levels of convergence near the surface.

In this case, the storm that will be analyzed developed near the Cheyenne Ridge, which is in an area that is historically considered favorable for terrain-influenced storms. The ridge most likely had a hand in changing the local wind field, which would support the formation of the outflow boundary and contribute to increased values of surface vorticity. The interaction may have strengthened the updraft speed and enhanced low-level rotation once the cell collided with the ridge, which lines up with other examples of ridge-influenced storm development. The influence of the Cheyenne Ridge provides a unique mesoscale forcing mechanism that lines up with radar evidence of boundary collisions and tornadogenesis.

4. Satellite Analysis

4.1.1 Water Vapor Imagery

The first category of imagery analyzed was the lower-level water vapor imagery, ABI band 10. For the satellite application, water vapor imagery is used to look at the amount and movement of moisture in the lower-upper levels. The wavelength of this imagery ranges from 7.3 in the lower levels to 6.2 in the upper levels in micrometers. While analyzing these images, it is important to know that dark and warm colors are indicative of moisture, while light and cold colors are indicative of higher moisture levels, which is also a good way to observe motion. WV images are also the framework for troughs and ridges, jet streams, and pockets of strong vorticity before drawing in the streamlines or isobars.

Beginning at 2151 UTC, there is a storm north of the focus storm, and it is reasonable to assume that this storm, along with the influence of the Cheyenne Ridge, was able to provide the necessary forcing for this storm to develop. During this same time, the very beginnings of convection began along the Colorado/Wyoming border as cumulus began to form in the late afternoon. Despite this convection, there appear to be no large-scale forcing mechanisms that are tied to this storm, at least in the lower levels of the atmosphere. Surrounding the beginning convection around 2200 UTC are values of much drier and stable air, which further enforces that this storm was heavily terrain influenced. Moving forward to 22:30 UTC, the storm of focus is rapidly developing, and higher values of moisture are observed as the storm continues its development. At the time of the tornado, there were prominent overshooting tops with visible shadows (Fig. 2), indicating that the updraft was very strong and rose high into the atmosphere. After the tornado lifted, the overshooting tops dissipated a bit, but they are seen later in the imagery (Fig. 1), which indicates the presence of the storm producing multiple strong updrafts after the focus event was concluded.

In summary, the water vapor imagery shows that this storm was not heavily under the influence of a larger scale system, but instead had local terrain and mesoscale influences, such as the Cheyenne Ridge. The overshooting tops and upper-level divergence further tell that the storm had enough support and forcing mechanisms to grow and produce a tornado.

4.2.1 Visible Imagery

The second utilized imagery in this analysis is ABI Band 1, which is known as Visible Blue. This band is efficient in monitoring cloud development in the daytime with a 0.5km spatial resolution. This band captures visible blue light, which is at the long wavelength end of the visible spectrum. This makes it efficient in seeing small-scale cloud features, shadows, and initiation, which can help determine severe convection by seeing overshooting tops and updraft activity.

Following the water vapor imagery around 22 UTC, the beginning of convection associated with this storm is observed along the Colorado-Wyoming border, which is in line with the convection seen in the previous imagery. In the late afternoon (22-2230Z), the clouds begin to take on a “lumpier” or “bubbling” appearance along with shadowing, which is indicative of the formation of overshooting tops and convection (Fig. 4). As the day continued, the storm expanded in all directions as it continued to work with the influence of the storm located to the north of the focus storm. The vertical growth of this storm shows that these parcels were rising efficiently, which is aligned with the lower LCLs mentioned in earlier analysis. East of the focus storm, in southeastern Nebraska, gravity waves can be seen forming. These gravity waves can be indicative of unstable air moving into a stable atmosphere, and as the air rises, gravity pulls it back down, as it cannot stay there indefinitely. These waves signal an interaction between stable and unstable environments, which would have a hand in convection and storm development. As the evening moved on, the storm continued to expand outward, continuing to produce storms with the influence of the storm to the north. The visible frame shows a persistence of bubbling cloud textures through all frames, which is indicative of a strong, lasting updraft, especially with the influence from the terrain as well as boundary interactions.

The visible imagery further reinforces the findings in the water vapor imagery. It provided clear and visual evidence of the vertical development through the “bubbling” cloud tops and the shadows formed from them. The gravity waves to the east of the storm show the interaction between stable and unstable environments and how they contributed to the environment.

4.3.1 Infrared Imagery

The final imagery utilized for analysis is the ABI 13 Band, which is known as the Clean Longwave Infrared Window. This window specifically is not as affected by water vapor values, which means that it provides a clearer and more concise view of could tops on the infrared (IR) spectrum. This specific band is efficient in seeing cold cloud tops, which are consistent with severe weather, and they show as a red/orange color. When cloud tops are colder, this means they are reaching higher in the atmosphere, which can mean stronger or strengthening updrafts within storms.

Shortly after convection on this storm began, a feature known as a “v-notch” can be seen at the top of the developing storm. This signature indicates a warm/cold couplet which can be indicative of a severe, even tornadic, storm. The actual “v” shape itself is very cold, which is usually connected to an overshooting top, where the storm’s updraft makes it to the lower stratosphere. The warmer air that trails the “v” formation is formed by the mixing of stratospheric air, which is relatively warm. The tails of the “v” formation are in the direction of the anvil, while the shape itself points in the direction of the updraft (Fig. 7). As time moves forward (22-23Z), this signature stays present and gets larger with time as the cell expands. At the time of the tornado (23Z), there was a very deep red shade over the area of the updraft, which indicates that the clouds in this region were extremely cold, meaning that the overshooting tops may very well be past the tropopause. After the tornado lifted, the deep shade of red lightened marginally, but in the frames following, the deep red shade came back, which indicates that updrafts may be present once again (Fig. 6). The growth of this signature is further proof of how strong the updraft was on this storm and why it continued to produce other storms after the focus event.

The infrared imagery helped to provide some of the strongest evidence of the intensity of the storm, with multiple instances of overshooting top signatures, v-notches, and cold temperatures that are well into the stratosphere. The signals before and after the tornado indicate that this storm was capable of producing repeated strong updrafts, which sustained the formation of multiple tornadoes.

Overall, the satellite frames show that this storm developed and intensified primarily due to mesoscale and terrain influences, rather than strong synoptic forcing. All of the satellite images show evidence of strong, persistent updrafts that supported the strengthening of the storm. WV imagery showed localized areas of increased moisture values and upper level divergence, while visible imagery showed rapid upward movement through the bubbling cloud tops and the shadowing that was associated with this storm. IR showed the strongest evidence of the intensity of the storm with consistent cold cloud top temperatures and a pronounced v-notch signature, which further supports the strong and sustained updrafts within the storm.

5. Radar Analysis

5.1 Base Reflectivity

This radar analysis will start with base reflectivity images sourcing from both the Cheyenne, WY (KCYS) Radar and the Denver/Boulder, CO (KFTG) Radar, which provided imagery across northeastern CO and southeastern WY. In this analysis, all radar and derived radar products had a tilt of 0.5° throughout all parts of this section. For all products, loops and still images were captured to allow for the analysis of storm initiation and convective development for this specific event. Following base reflectivity, base velocity, correlation coefficient (CC), spectrum width (SW), probability of severe hail (POSH), along with other will be used to form the analysis of this specific storm.

Beginning around 22:40z on May 23, there is a cell moving southeastward on base reflectivity, initiating northeast of Keota, CO. South and slightly eastward of the observed cell, there is an outflow boundary (OFB) that the cell is latched onto as it moves southeasterly. During the focus tornadogenesis, this storm had a pronounced hook echo (Figure 8), and the reflectivity values around 23:30Z-23:56Z were measured around 60-70dBZ with some higher values of 80dBZ. These higher values of reflectivity are usually associated with the presence of rain or hail, and these higher values look to be consistent with the location of the rear flank downdraft (RFD). Observations beginning at 23:09Z confirmed that there was a tornado produced during this time frame (Figure X) that was moving southeastward near Merino, CO. During the same period, hail markers around the same area indicated hail as large as 2.5” to accompany the tornado reports. At 23:38Z, radar imagery displayed dBZ values as high as 75dBZ, which can be an indicator of a strong updraft. During this time, tornadogenesis began (Figure 8), and a tornado was produced, or fully formed, just moments later. This tornado would go on to stay “on the ground” until 23:56Z, where it began to dissipate and rope out. The initiation of this storm likely had to do with local terrain influences, such as the Cheyenne Ridge, which is located on the border of WY and CO. This terrain feature increased values of vorticity near the surface, which was beneficial to the development of this storm, integrated with the thermodynamic indices in place. After the initial rotation began to dissipate, the focus shifted to a storm forming south of this weakening rotation. Along with the OFB, there is another observed OFB that moved slowly northward, which eventually collided with the cell of focus. The inflow of the cell collided with the cooler and denser air associated with the OFB, which created a zone of strong convergence near the surface. This forced air upward, which intensified the updraft in the lower levels of the storm. After the collision of the cell with the OFB, rotation began to tighten and strengthen near the southwestern flank of the storm, and the first observation of a brief tornado on the southern storm was sent at 0:11Z. Later frames on GR2 Analyst show evidence of a tightening hook echo northeast of Platner, CO, which would go on to produce another tornado later in the evening.

5.2 Dual Pol Products

The next section of this analysis will remain focused on certain derived products. The first part will focus on velocity products, both base (BV) and storm relative (SRV), to analyze factors such as inbound and outbound winds, as well as velocity couplets within the storm. Other products that will be analyzed are correlation coefficient (CC), spectrum width (SW), echo tops (ET), and probability of severe hail (POSH).

The analyzed radar images will focus on the specific time frame of 22:45Z to 23:56Z, which lines up with the conclusion of the focus storm. This time frame is the most effective in showing the development and growth of the storm, as well as finding when the storm reached its peak intensity. The radar scans will be shown on 4-panel figures, showing BV, CC, SW, and POSH, with some frames being singled out to ensure correct analysis (GRL2). During section 5.1, base reflectivity was analyzed, and the cell of interest looked like most “textbook” supercells, with a notch, a well defined rear flank downdraft (RFD), and a well defined forward flanking downdraft (FFD). Any differences in development can most likely be attributed to the unique terrain and environment associated with the location of the focus storm.

Base Velocity (BV) is a WSR-88D product that depicts a full 360 sweep of echo intensity data, and it is available for every elevation angle in a volume scan. When looking at BV during the beginning of the scan, the cell is moving southeastward off the Cheyenne Ridge toward Willard, CO. There are observed areas of broad rotation that tighten as the storm continues its path toward Atwood, CO. At 23:09 UTC, the first tornado report was observed on this cell, and the tornado was said to last around 5 minutes before dissipating. Rotation broadened after the dissipation, but tightened shortly after around 23:32Z, which is minutes before the Akron tornado formed. The velocity couplet for this tornado can be seen on Figure 12, indicating a focused area of rotation. This tight velocity couplet can be inferred as a good indicator for rotation in the storm, as well as indicating a possible tornado, but this product is not capable of “knowing” if a tornado is happening. Despite the uncertainty, the velocity couplet is usually a strong sign of tornadic activity.

Akin to BV, SRV is a velocity product that takes a 50km x 50km region of storm relative mean radial velocity on a point where it can be specified in a 230km radius of the radar. This product will default to the motion of the storm closed to the product center and is a good way of analyzing how wind interacts with the storm structure. Combined with BV, this could be a good indicator of severe storms and storms with strong rotation. Almost identical to BV, SRV showed a strong velocity couplet that lines up directly with the time of the tornado formation. There is also evidence of some possible aliasing (Fig. 13), which is when the winds appear to change or “flip” direction due to the strong velocity values. This aliasing is usually indicative of very strong inbound or outbound flow within the couplet, which supports the presence of strong LL rotation.

Similar to Base Reflectivity, there is an area of tightening rotation near the “hook” of the storm, which can be indicative of tornadic activity (Fig. 13). Figure 14 shows a cross section of SRV during peak intensity, and the "tilt" in the updraft can be visualized by viewing this product. The tilt in the updraft is a result of wind shear, as the lower part of the updraft is displaced from the upper part of the updraft, further supporting tornadic development (Fig. 14).

Correlation Coefficient (CC) measures how similarly radar scans act in the horizontal and vertical dimensions, which makes it useful for detecting debris and hydrometeors. During this event, CC displayed a brief drop beneath the updraft and mesocyclone (Fig. 15) during the time of the tornado. Although a bit chaotic, the drop in CC values was significant because it suggests the storm was lifting debris. In addition to any debris, the mixture of rain/hail within the hook region lowered the CC values, which is aligned with reported hail sizes and higher reflectivity values seen on Base Reflectivity. The drop in CC values confirms that there was strong vertical motion and low-level rotation present, which reinforces the radsr signatures associated with tornadogenesis. A cross section of this product also shows low values of CC higher than 20,000ft in the atmosphere, which is a strong indication of lofted debris from the storm (Fig. 17).

Spectrum Width (SW) provides a degree of velocity dispersion within the storm. Because SW represents the variability of radial velocity estimates within a single radar scan, this product becomes very helpful in finding areas of wind shear, turbulence, and overall chaotic motion. SW is used to find areas of turbulence associated with LL boundaries, mesocyclones, and regions of strong shear in supercells. During this event specifically, increased values of SW were associated with the northward-moving OFB, which interacted with the storm right before tornadogenesis. OFBs often have gradients of both wind direction and speed, and when an OFB collides with a storm, there can be elevated values of shear and chaotic motion (Fig. 17). This turbulence was depicted in the storm as it collided with the OFB, which is indicative of a changing low-level wind field with strong shear and chaotic motion. The increase in SW aligned with the strengthening velocity couplet, suggesting that there was boundary-influenced turbulence that contributed directly to the intensifying rotation in the lower levels. SW played a critical role in analyzing the storm's structure and provided valuable evidence that the storm was moving into an area with favorable conditions for tornadogenesis.

Echo Tops (ET) is a product that measures the height above ground of the center of the radar beam using a tilt or scan that contains the highest elevation where reflectivity values greater than 18.5dBZ can be detected. ETs seem to be consistent throughout most of the life cycle of the storm (around 50KFT), specifically after kicking off of the Cheyenne Ridge. A higher value of ETs usually indicates higher and colder clouds, which is a trademark of stronger storms. Since these values stayed consistent for most of the storm's life cycle, it can be inferred that this storm was strong throughout the entire time it was observed on radar, or at least during this specific time frame (Figure 18). Combining ET with POSH will be beneficial in providing a more put-together analysis of hail potential, because POSH quantifies the probability of severe hail, and Ets indicate the storm's vertical development and updraft strength. High values of both parameters, which are present in Figure 18 and 19, are likely to indicate mature storms that can produce large and damaging hail. Like combining POSH with ETs, combining it with base reflectivity will represent a relationship known as hailfall kinetic energy (HKE). It can be summarized as a temperature-weighted vertical integration of hail energy throughout the column of the storm.

Across all products, there is strong evidence of a well organized supercell that continued to strengthen as it interacted with mesoscale boundaries. Base reflectivity showed the storm's structure, containing a hook echo, strong core of increased reflectivity values, and RFD and FFD signatures. These features aligned directly with BV and SRV, both of which visualized a strengthening velocity couplet at the same time and location of tornadogenesis. The evidence of aliasing in these products further supported the presence of strong LL winds within the circulation of the storm. Drops in CC aligned with the velocity couplet and hook echo, which is indicative of lofting debris, while increases in SW values showed more chaotic flow as the storm interacted with outflow boundaries. ETs and POSH visualized consistent and deep convection with a high probability of severe hail, which matched the strong updraft signatures seen across other products that were analyzed.

6. Terrain Influence

For this event specifically, the Cheyenne Ridge has a large influence on the storm's initiation and organization. Radar and Satellite imagery from 2141Z and 2221Z (Fig. 30), shows that the developing cell stayed directly along the western edge of the ridge, where the southeasterly winds, modified by the upslope, ran into the elevated terrain. Within this area, low-level winds were redirected and quickly went upslope, which produced strong values of local convergence and shear, which are conditions that align with previously documented terrain-influenced storms. Over the 40-minute window shown in Figure 30, reflectivity values increased quickly as the cell grew, which is indicative of the storm being influenced by the localized areas of lift created along the ridge.

As the storm grew stronger and moved ESE off the ridge, its interaction with the OFB strengthened low level circulation even further. The sequence of terrain influenced convergence and boundary interaction is consistent with satellite and radar observations of overshooting tops and tightening rotation prior to tornadogenesis. Rather than being a background or less important influence, the Cheyenne Ridge had a large effect on the storm's development by modifying local wind profiles and creating localized zones of convergence.

7. Conclusion

The May 23, 2025 Akron, Colorado supercell shows how a tornadic storm can intensity and longevity in an environment where synoptic-scale forcing was weak, and the storm was instead relying on mesoscale processes and terrain-influenced changes to the low-level flow. Satellite imagery across water vapor, visible, and infrared bands consistently documented evidence of strong and persistent updrafts through bubbling cloud tops, cold overshooting tops, and localized regions of upper-level divergence, which are all strong signals that the storm was repeatedly able to strengthen, despite the limited amount of large-scale ascent present in the environment. Radar analysis strengthened these factors, with base reflectivity revealing a classic supercell structure and BV/SRV showing strengthening velocity couplets that aligned with the documented tornadogenesis. Drops in correlation coefficient provided strong evidence of periods of debris lofting, while increases in spectrum width showed increased chaotic motion and shear as the storm interacted with multiple outflow boundaries. These boundary collisions, combined with a saturated boundary layer and strengthening lower-level shear values, created a localized zone of enhanced rotation near the surface that supported tornadogenesis. An important factor influencing these processes was the Cheyenne Ridge, which changed surface wind profiles, increased localized convergence, and supplied initial lift that allowed the storm to mature as it moved off the terrain. All together, the multi-factor analysis shows that even without strong synoptic forcing, the overlay of terrain influences, mesoscale boundaries, and convective instability can create a favorable environment for supercells, tornado production, and large hail, which points out the importance of integrating satellite, radar, and surface and environmental data when evaluating storm potential.

*****In this analysis, Grammarly was utilized for citations, as well as to ensure correct grammar and clarity (Grammarly, Grammarly Inc., 2025). ChatGPT 5 was utilized to ensure correct grammar and structure, as well as to aid in finding references/sources that were later utilized in this analysis (OpenAI. ChatGPT. 28 Sept. 2025 version). In no way were either of these sources used to write any of the material in this analysis.*

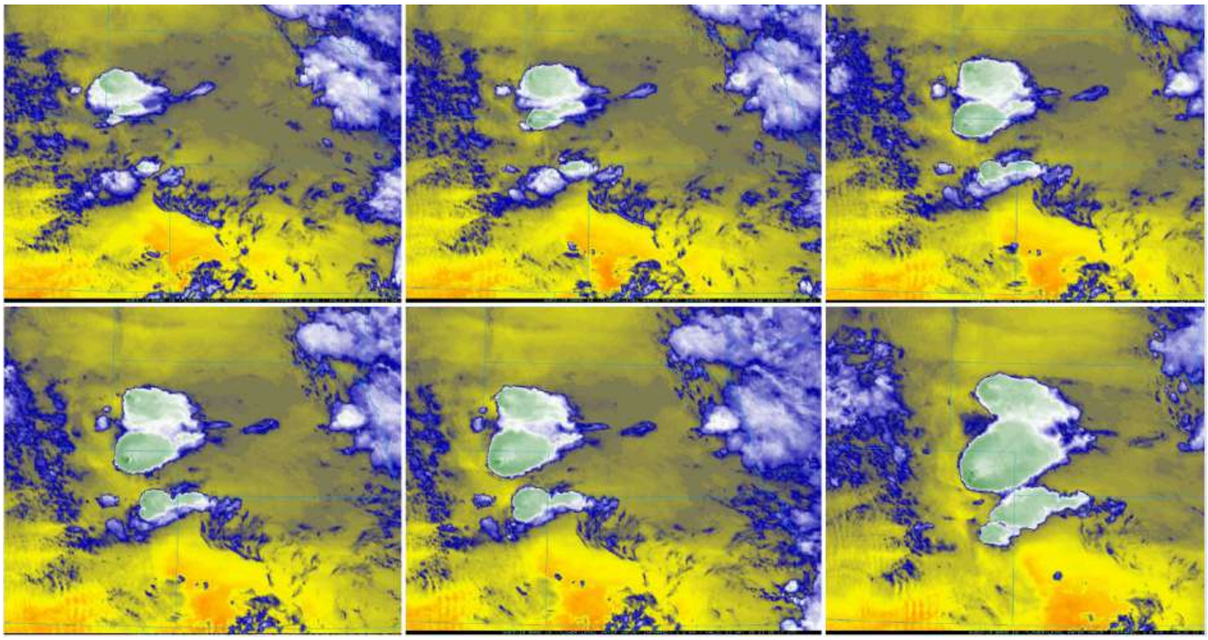

Fig. 1: Mid-Level Water Vapor frames of the Akron storm (2201z-2357z)

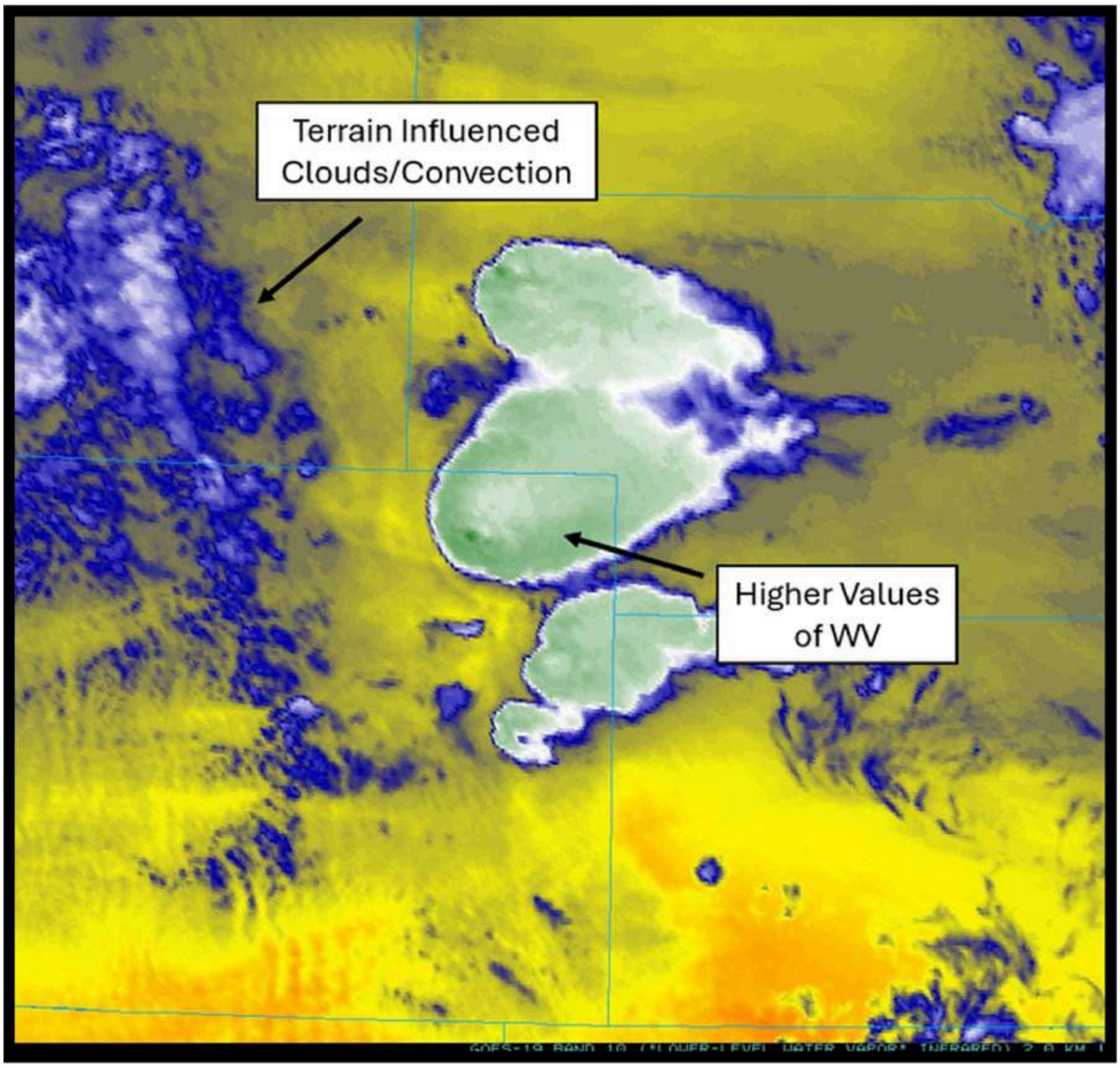

Fig. 2: Mid-Level Water Vapor values at the time of peak intensity (2337z)

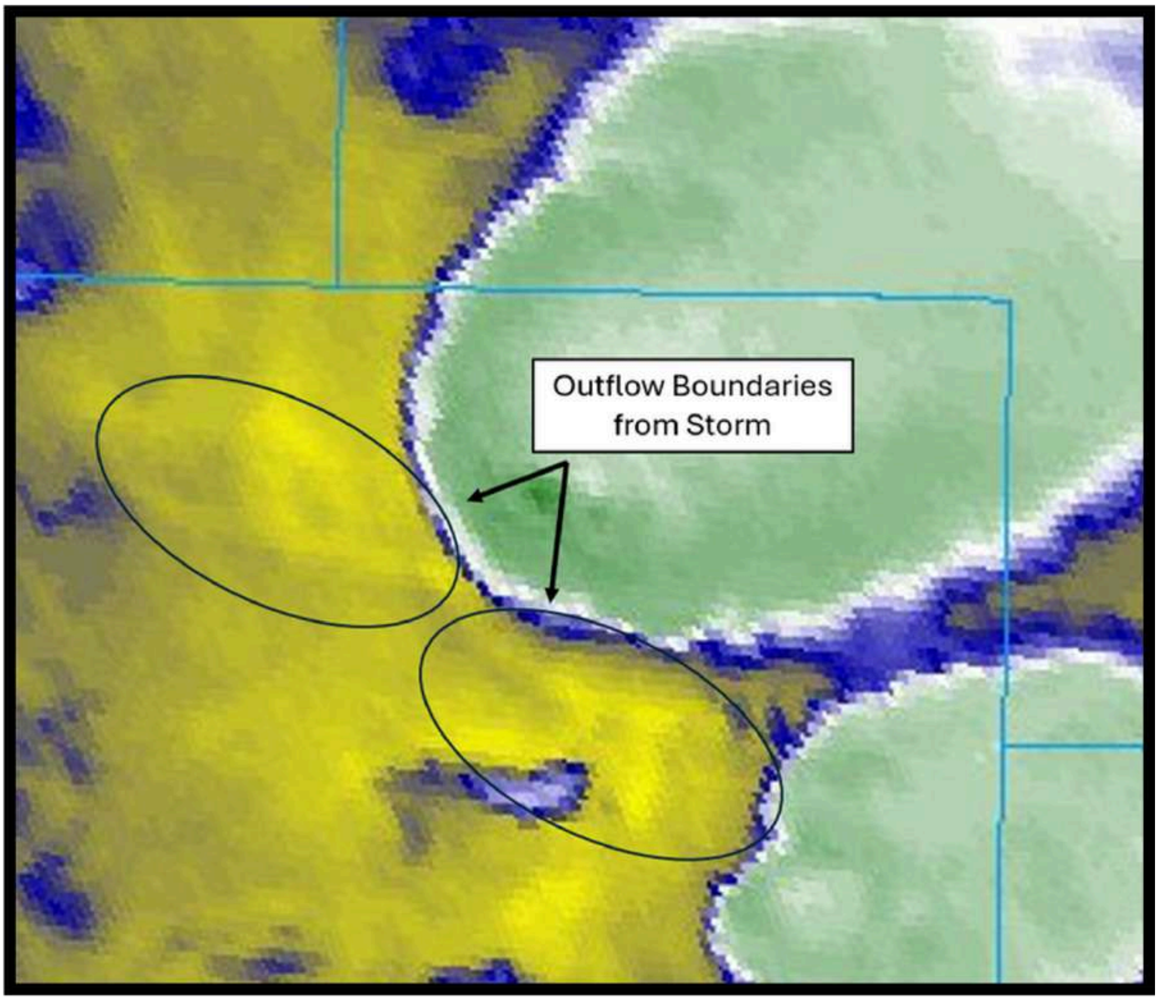

Fig. 3: Mid-Level Water Vapor values at the time of peak intensity (outflow boundaries highlighted) (2337z).

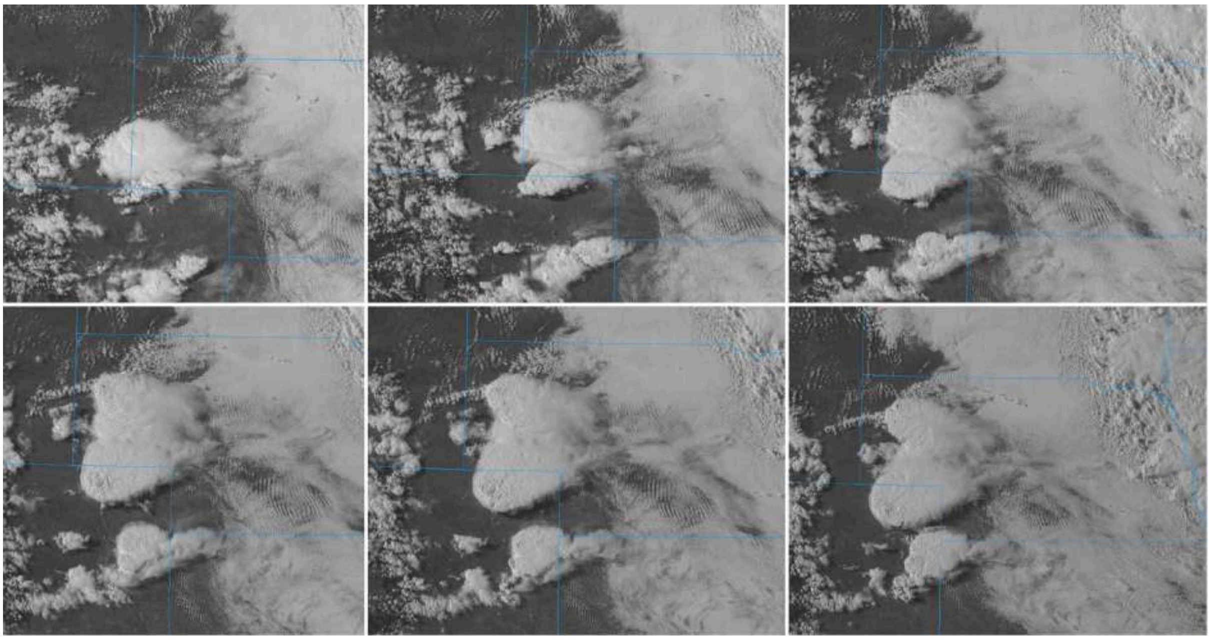

Fig. 4: Visible Blue frames of the Akron storm (2201z-2357z)

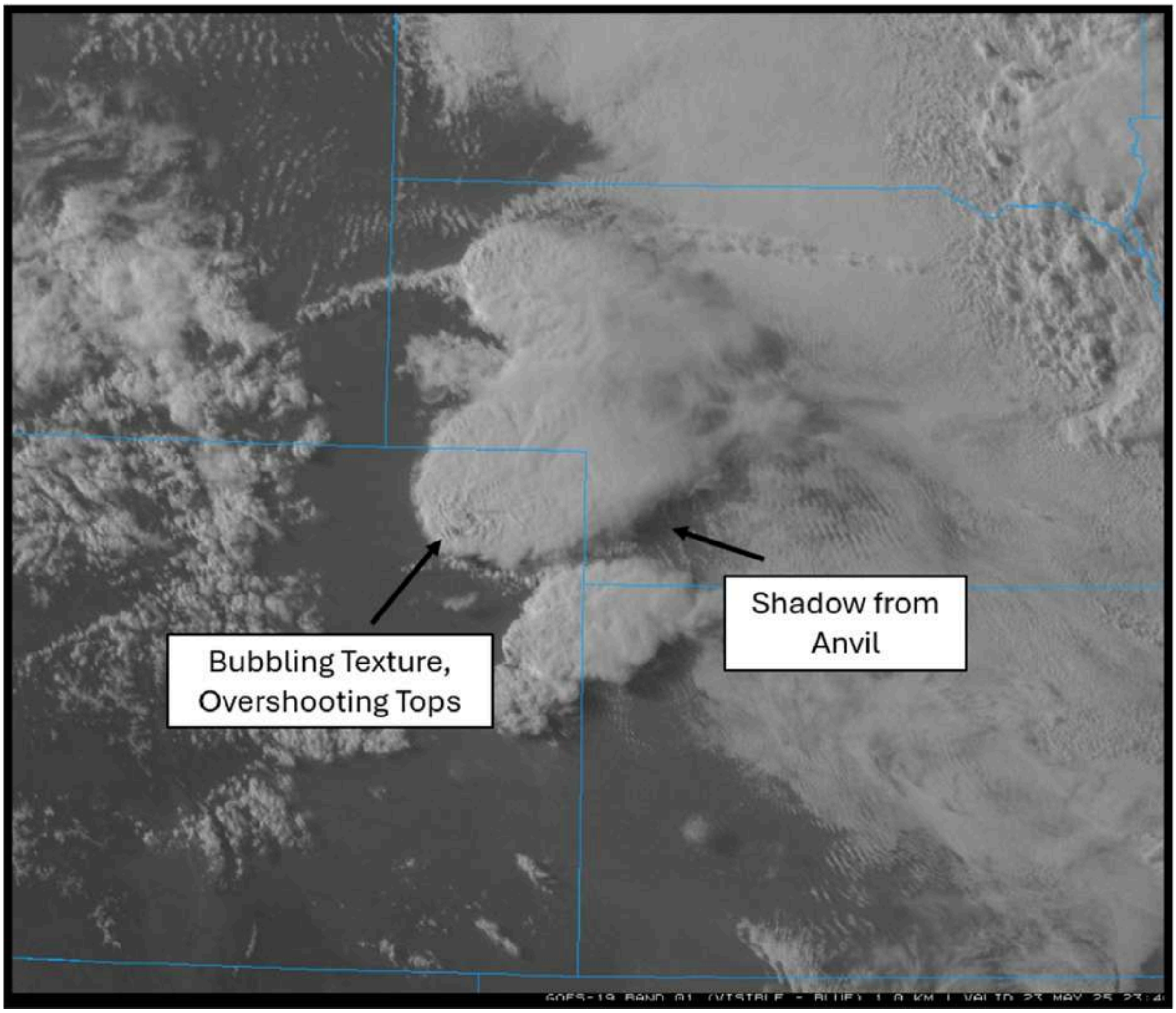

Fig. 5: Visible Blue frame from peak intensity (2337z)

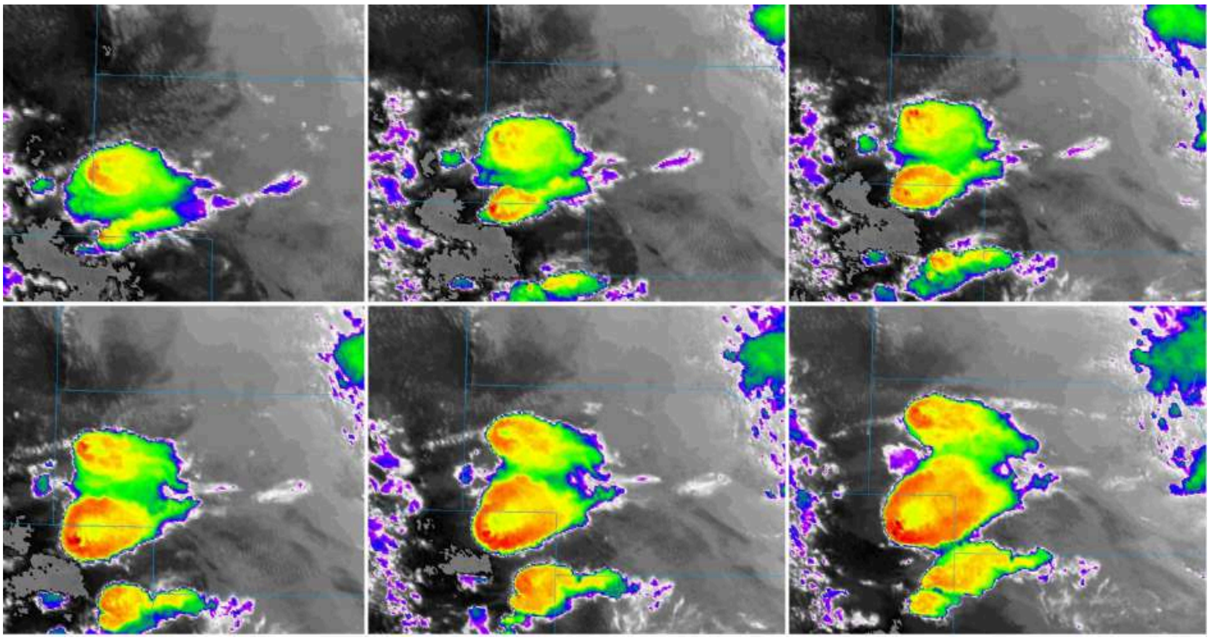

Fig. 6: Clean Infrared Imagery frames of the Akron storm (2201z-2357z)

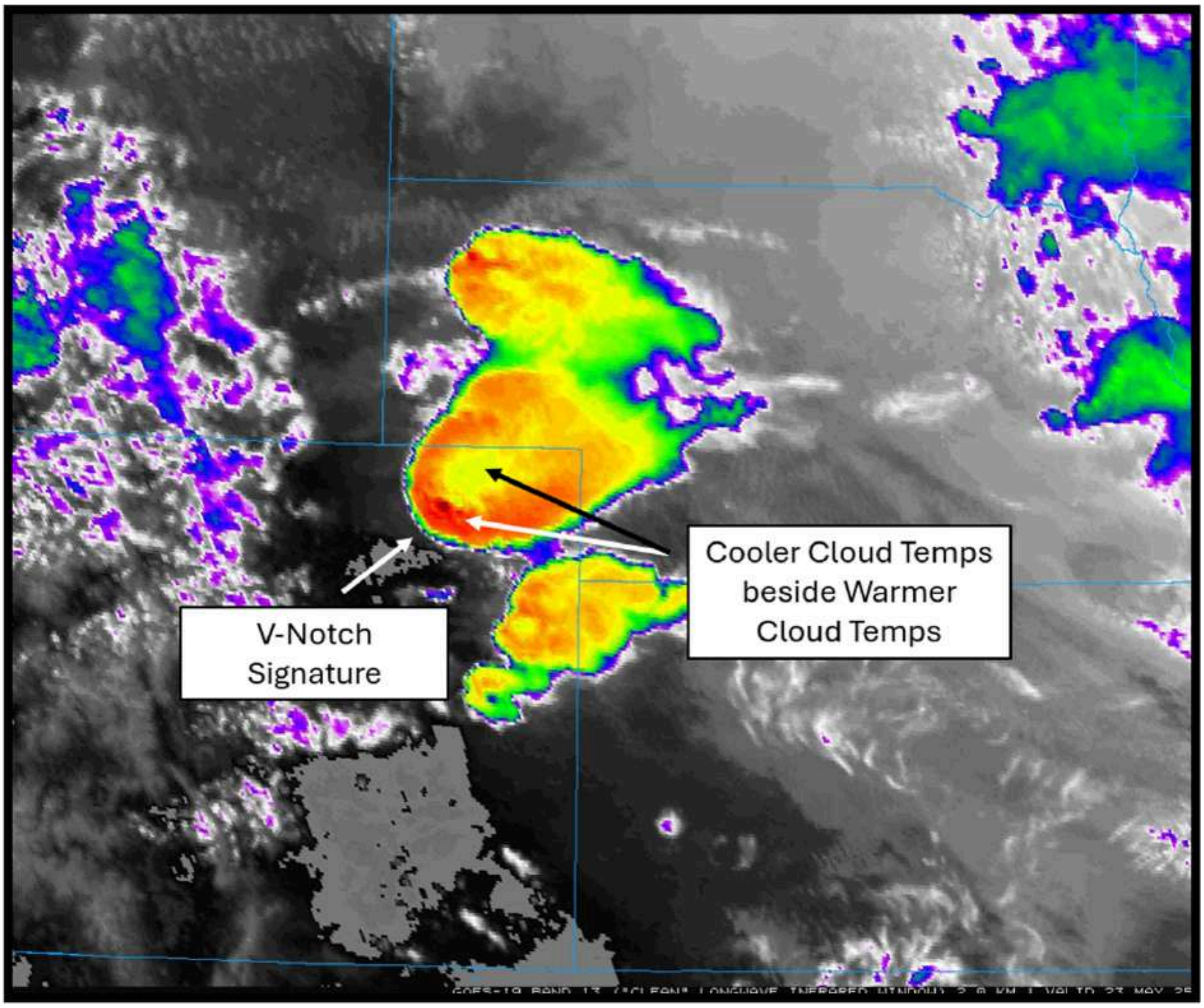

Fig. 7: Clean Infrared Imagery frame during peak intensity (2337z)

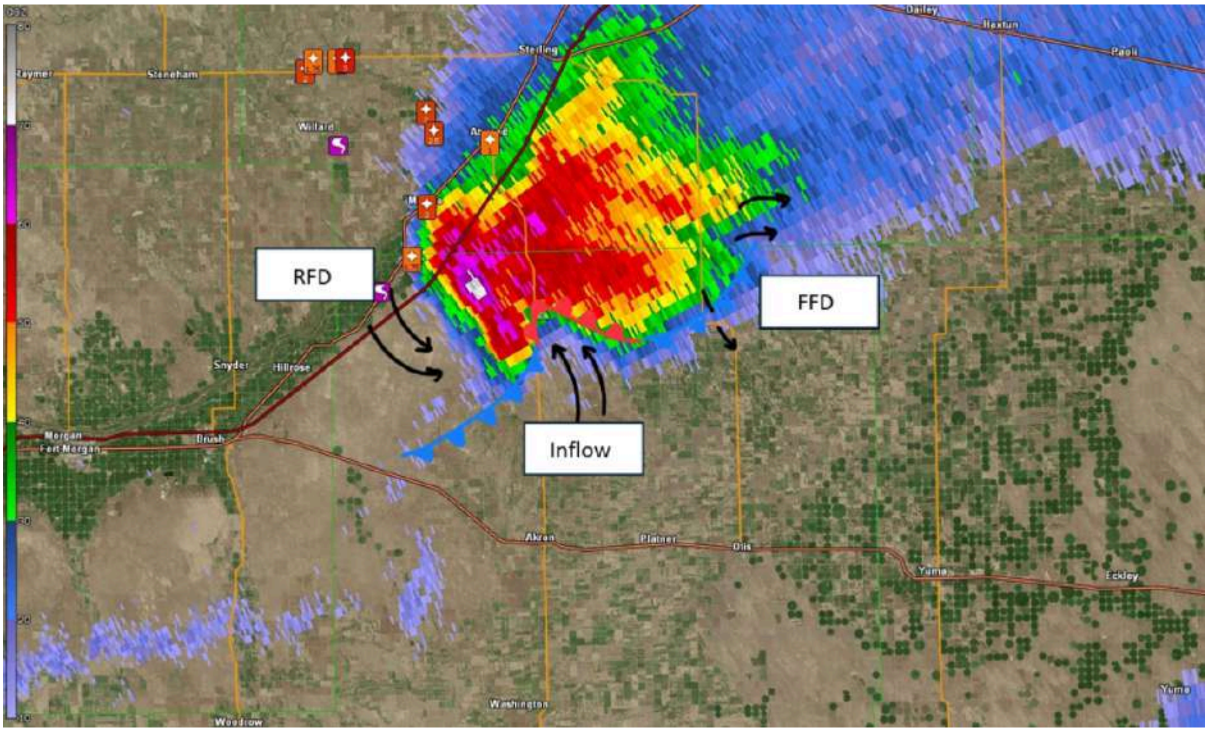

Figure 8: Base Reflectivity frame during peak intensity (2337z)

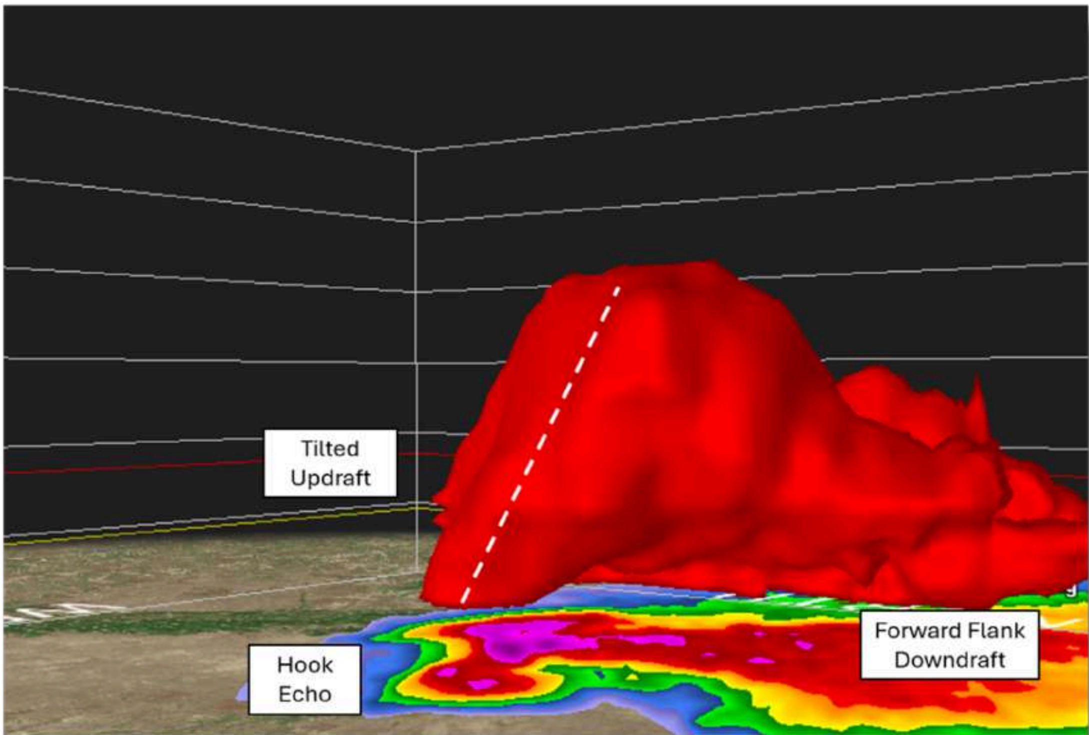

Figure 9: Isosurface Image (Base Reflectivity) showing tilted updraft (2337z)

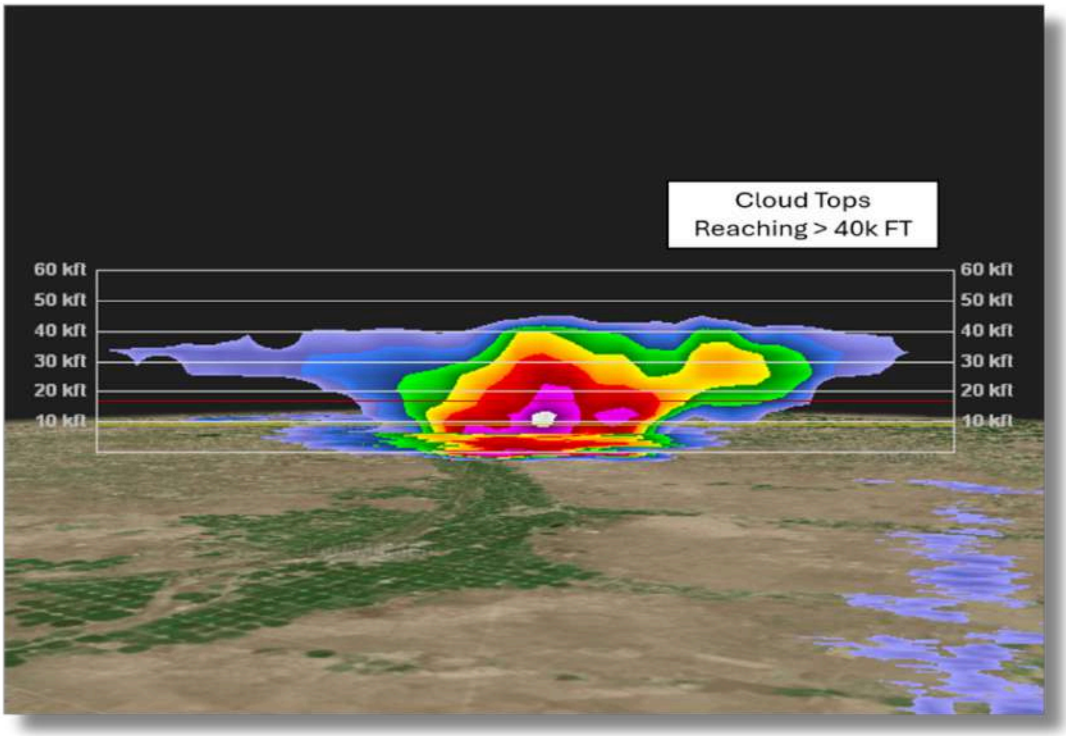

Figure 10: Base Reflectivity cross section during peak intensity (2337z)

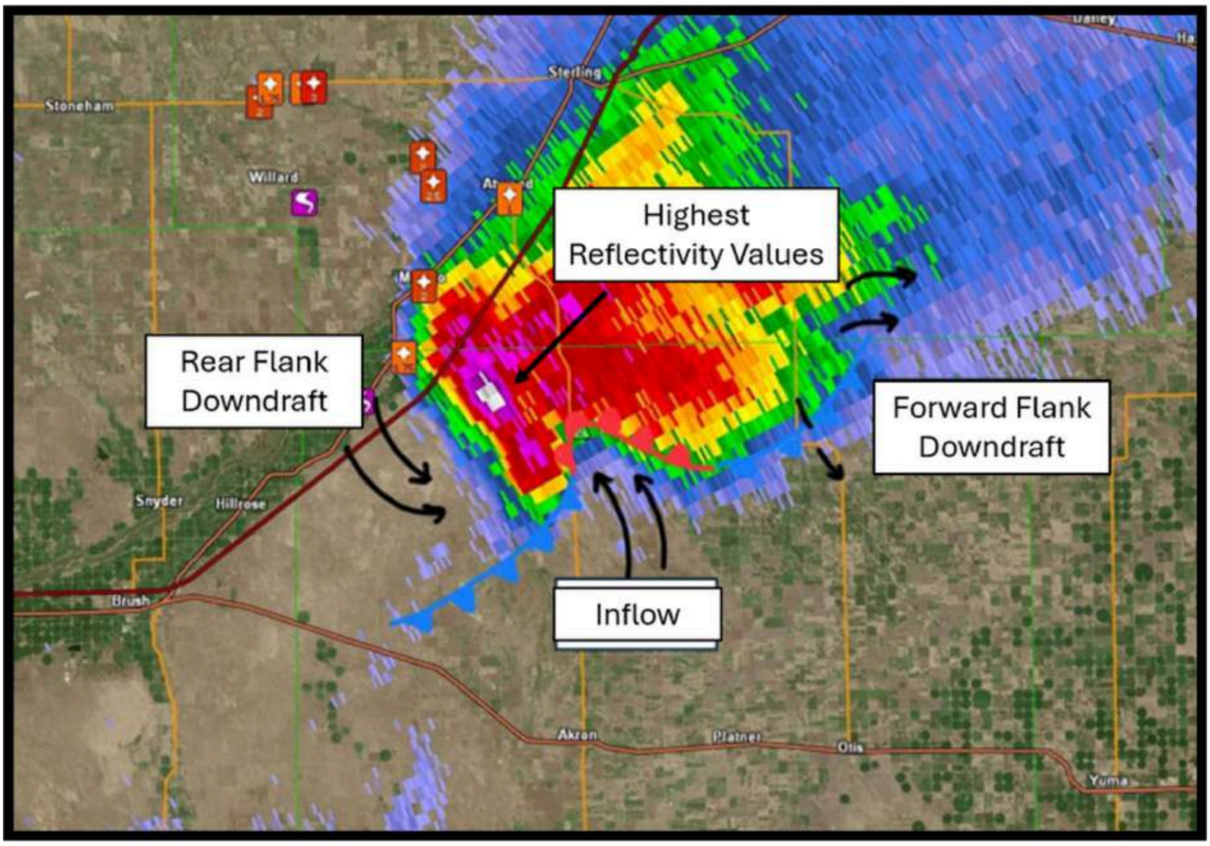

Figure 11: Base Reflectivity frame during peak intensity with annotations (2337Z)

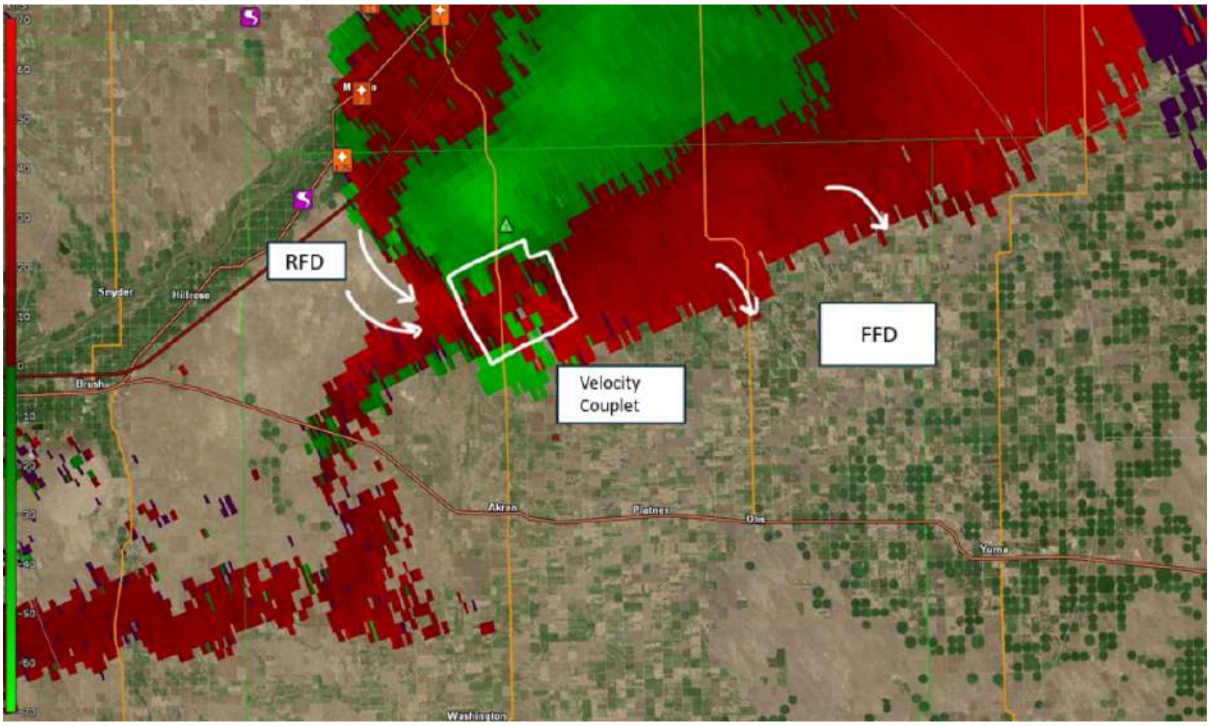

Figure 12: Base Velocity frame during peak intensity (2337z)

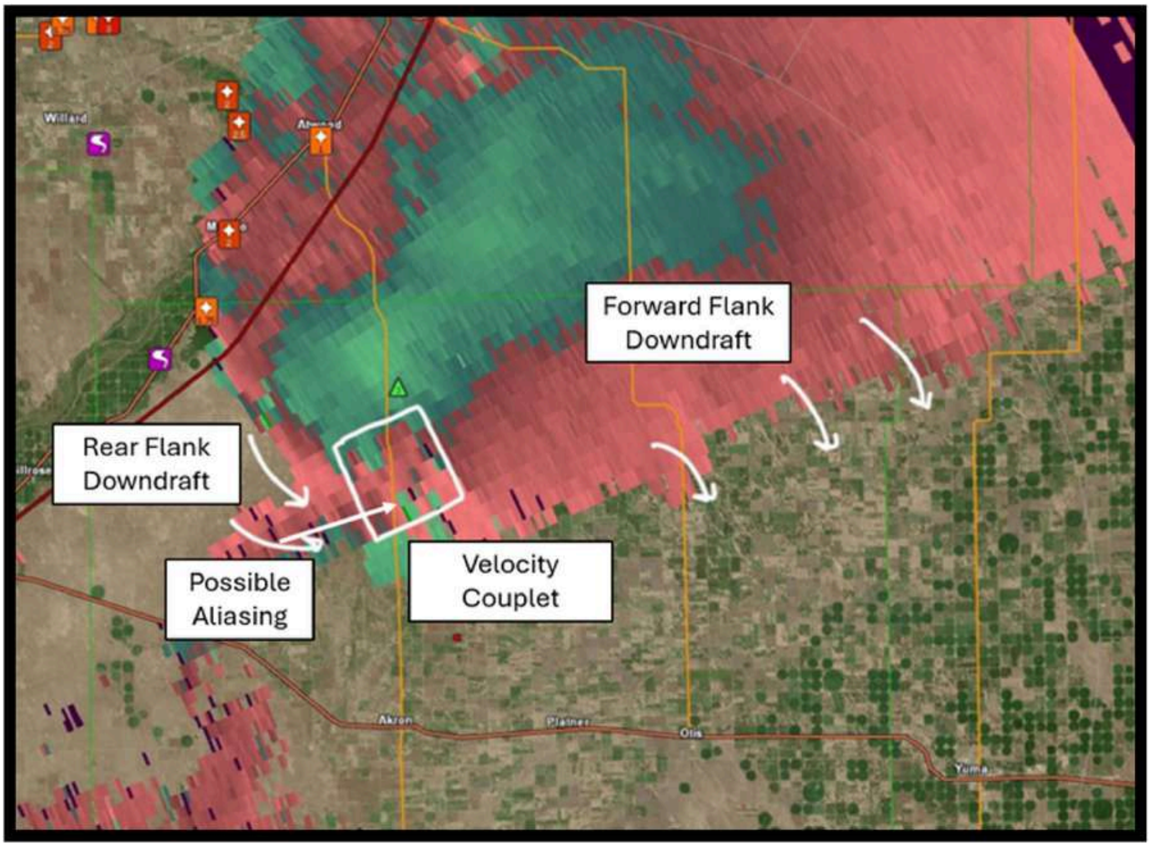

Figure 13: Storm Relative Velocity frame during peak intensity (2337z)

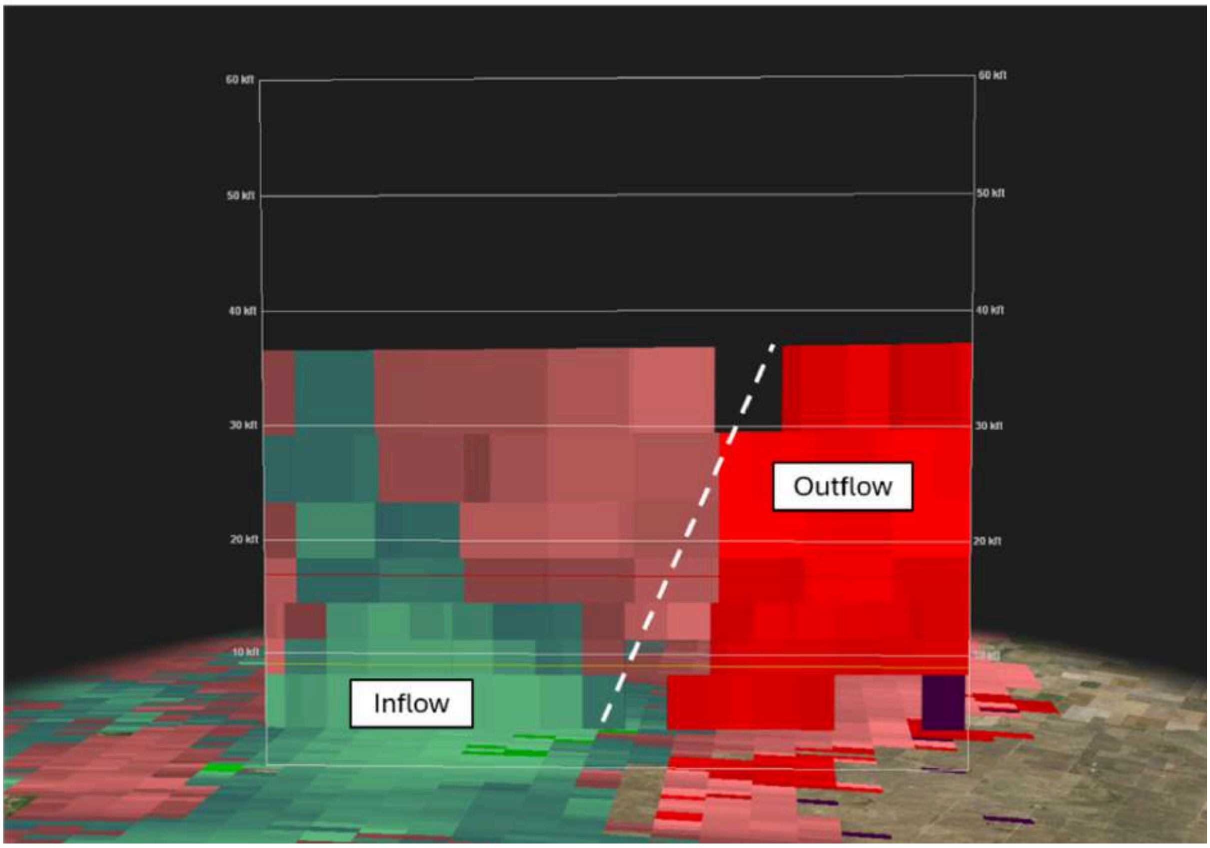

Figure 14: Storm Relative Velocity cross section during peak intensity (2337z)

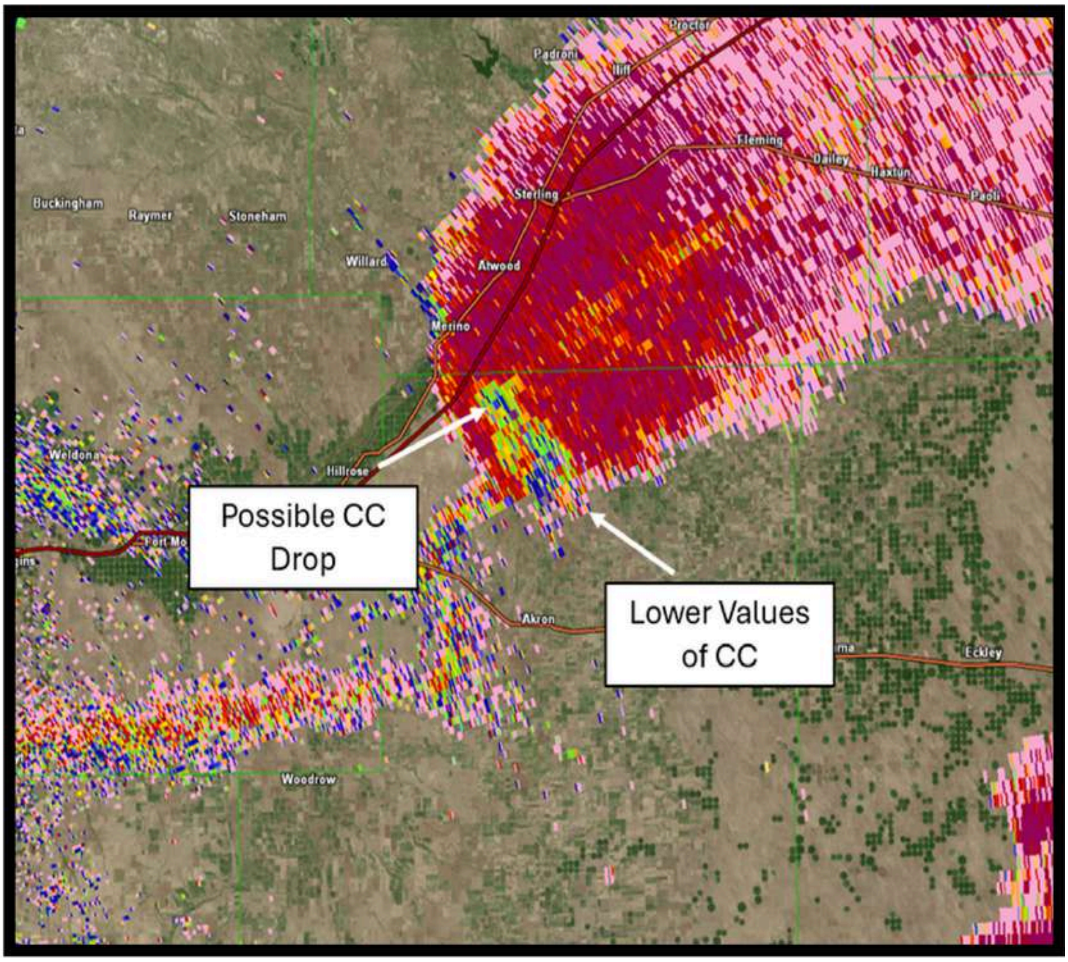

Figure 15: Correlation Coefficient frame during peak intensity (2337z)

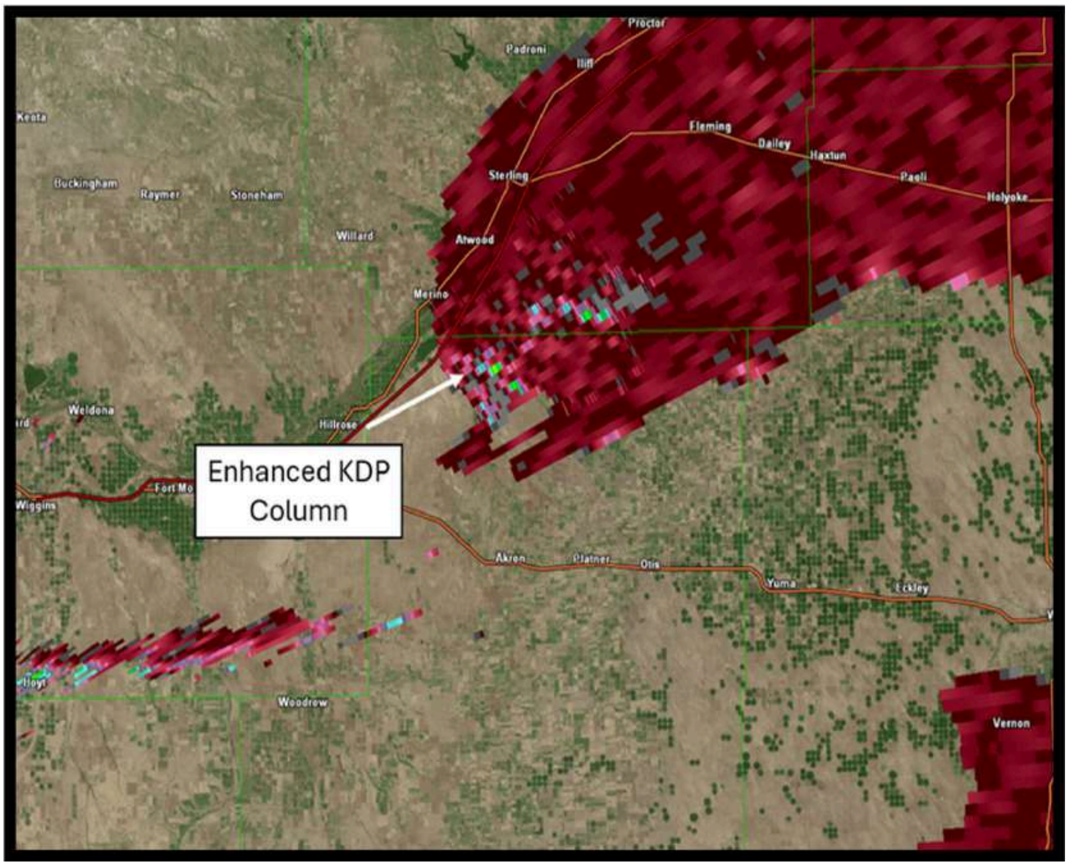

Figure 16: Specific Differential Phase frame during peak intensity (2337z)

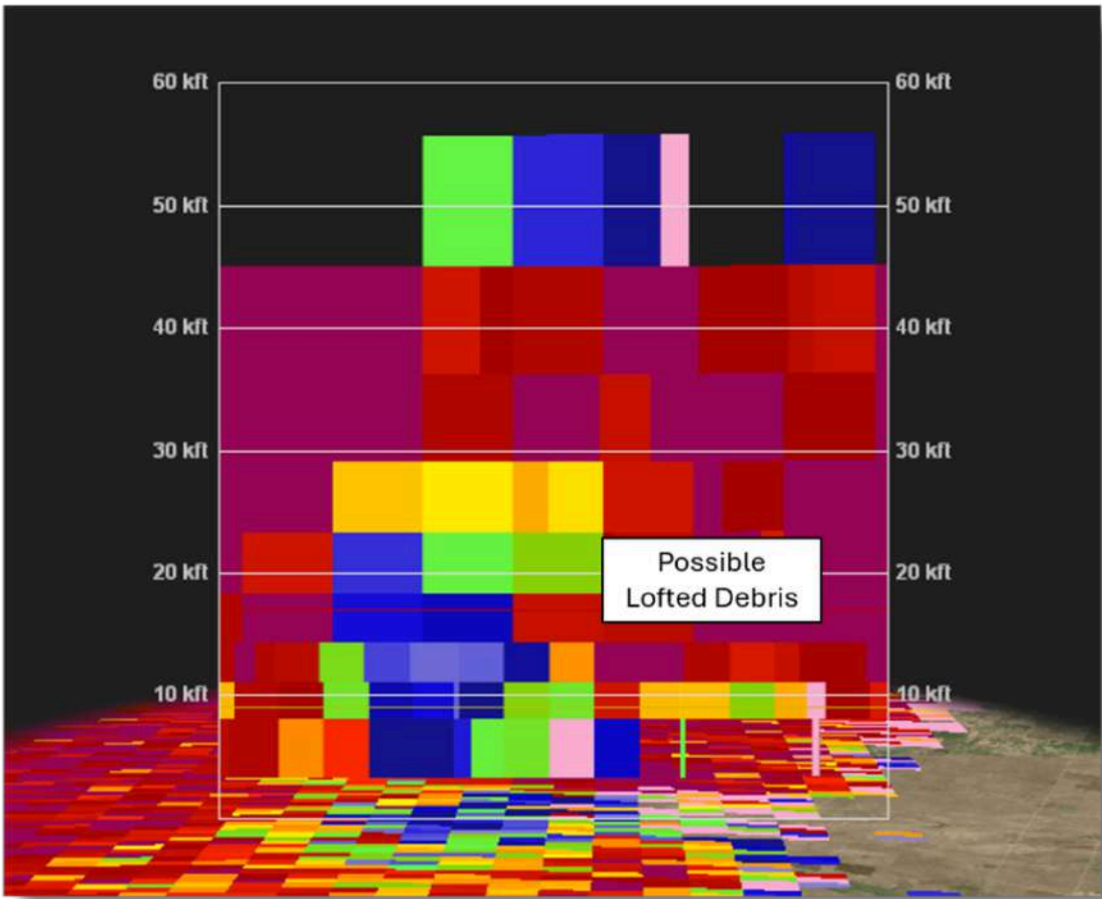

Figure 17: Correlation Coefficient cross section during peak intensity (2337z)

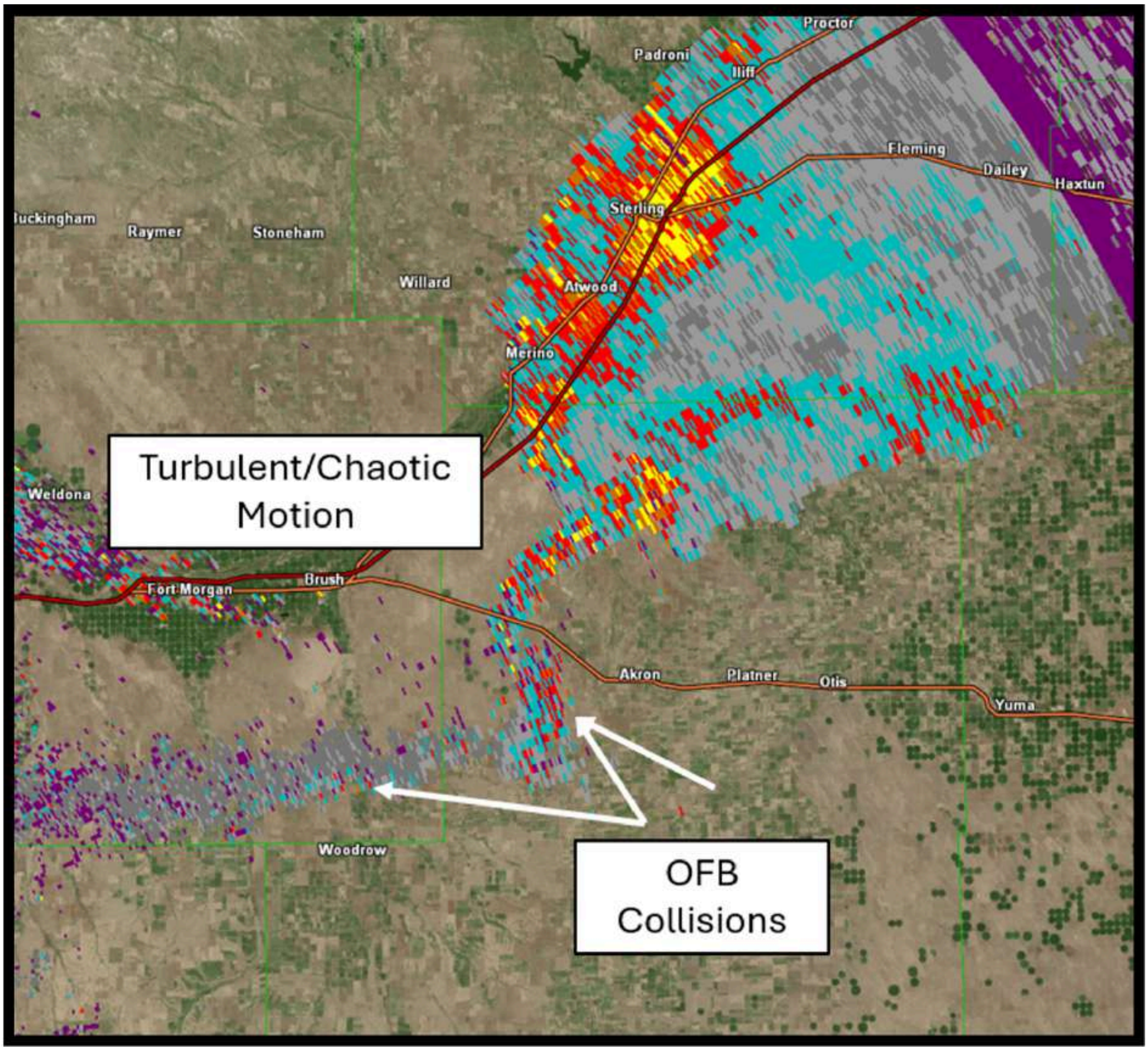

Figure 17: Spectrum Width frame during peak intensity (2337z)

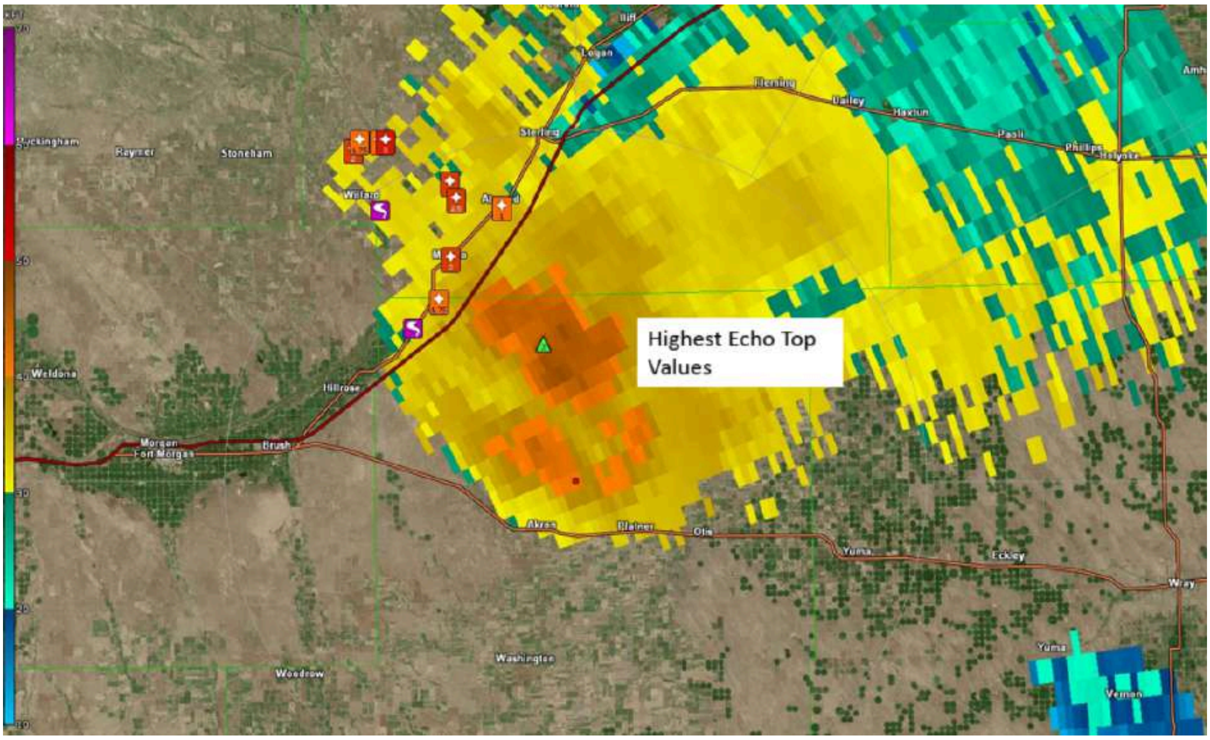

Figure 18: Echo Top values frame during peak intensity (2337z)

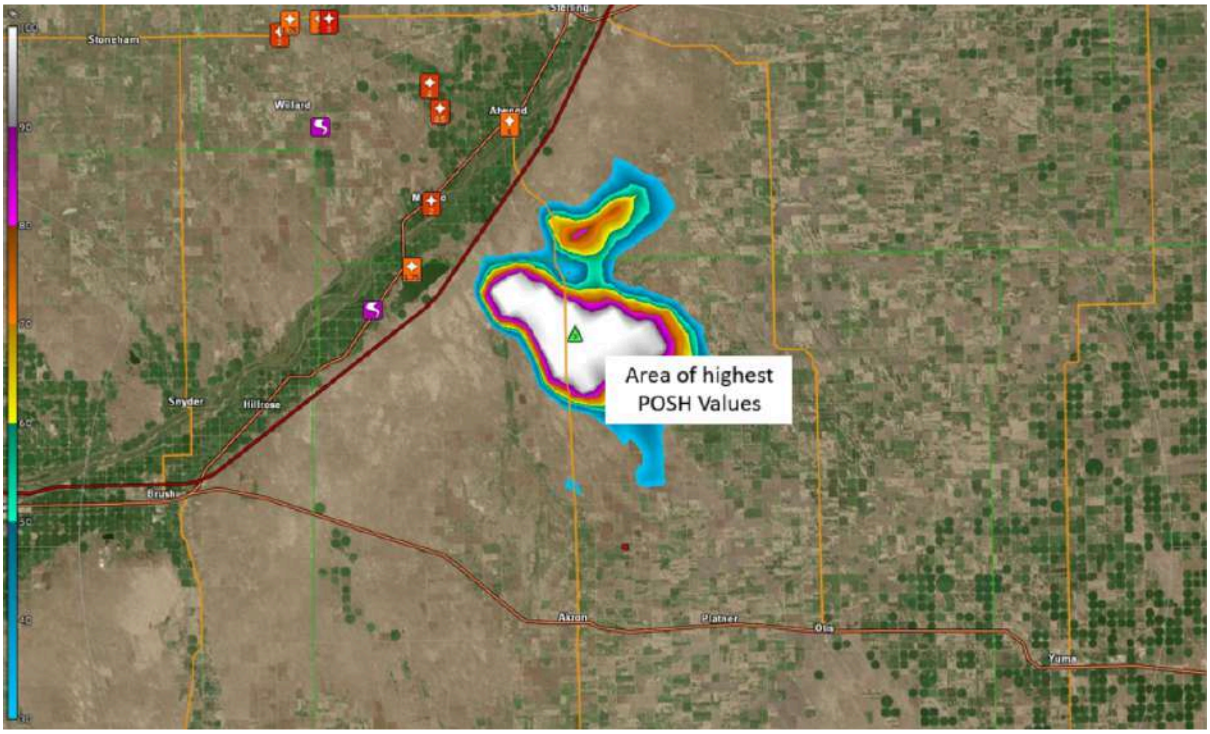

Figure 19: Probability of Severe Hail values frame during peak intensity (2337z)

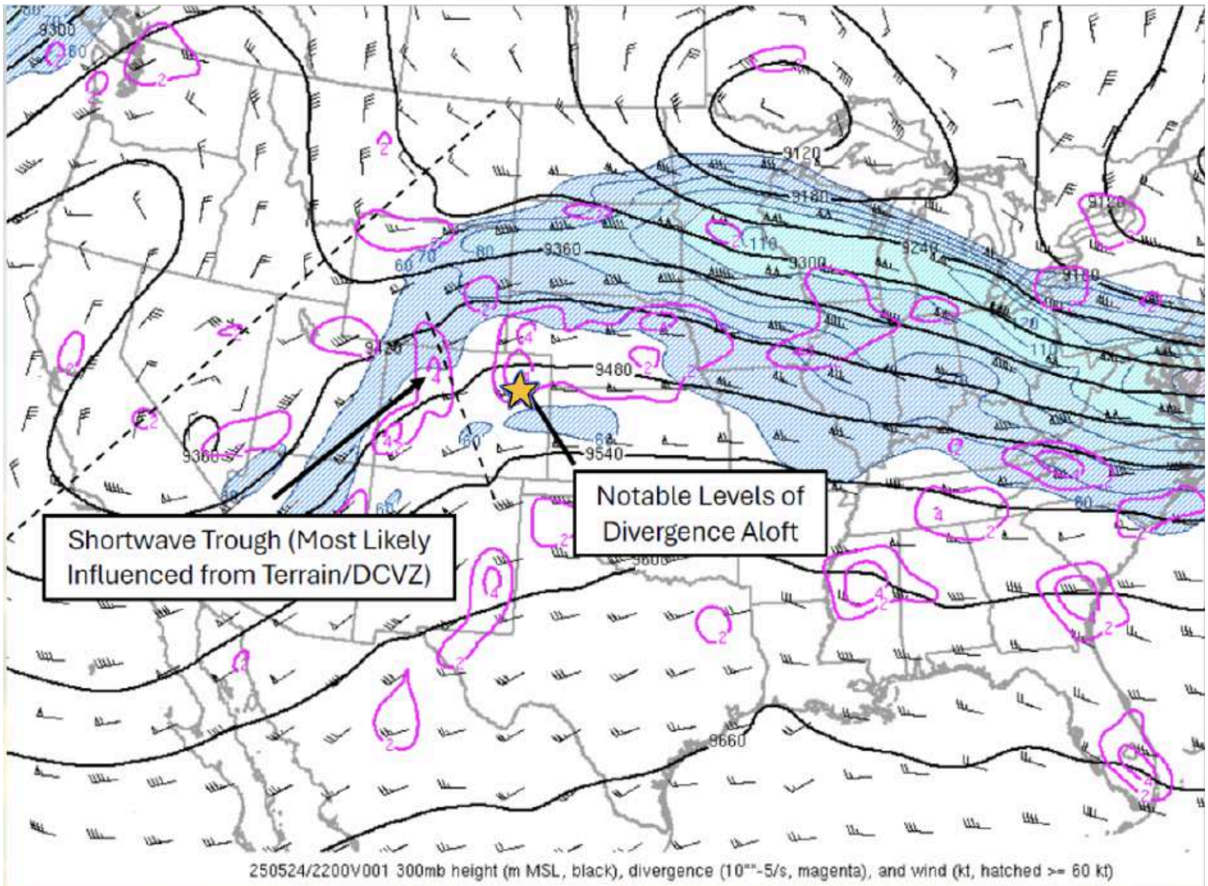

Figure 20: 300mb Heights, Winds and Divergence (2200z)

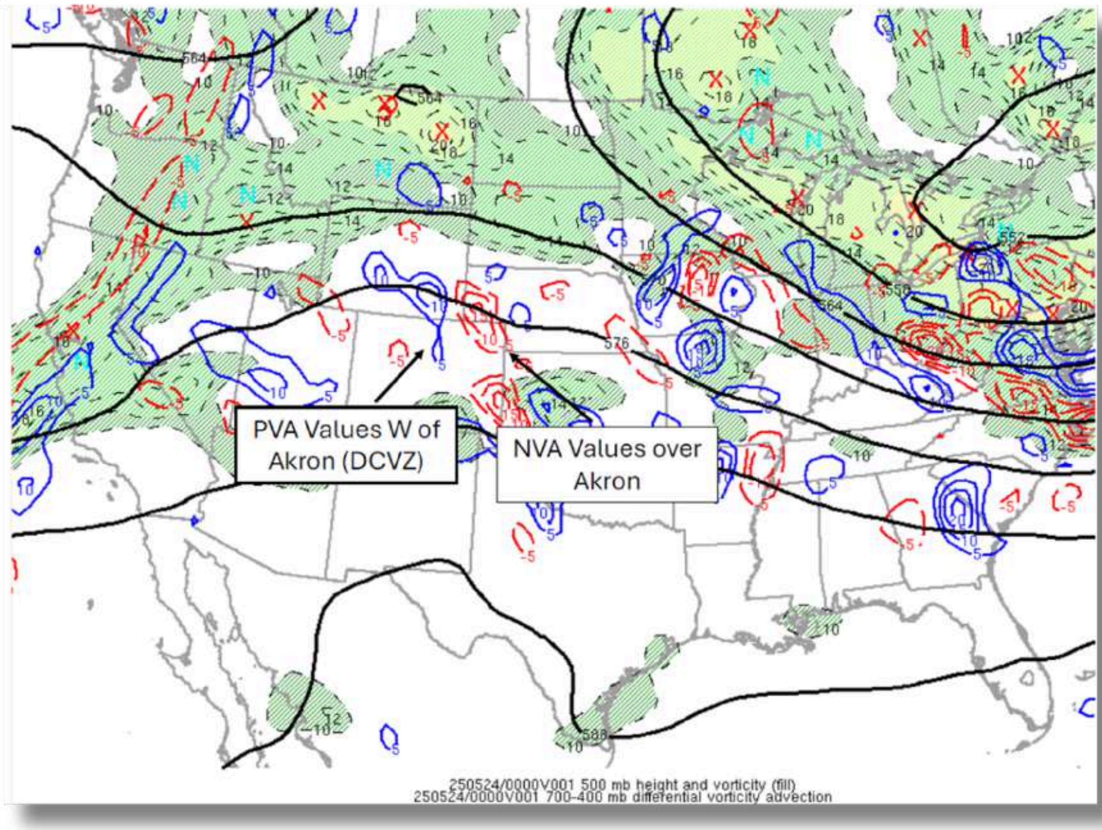

Figure 21: 500mb Height and Vorticity, 700-400mb Differential Vorticity Advection (0000z)

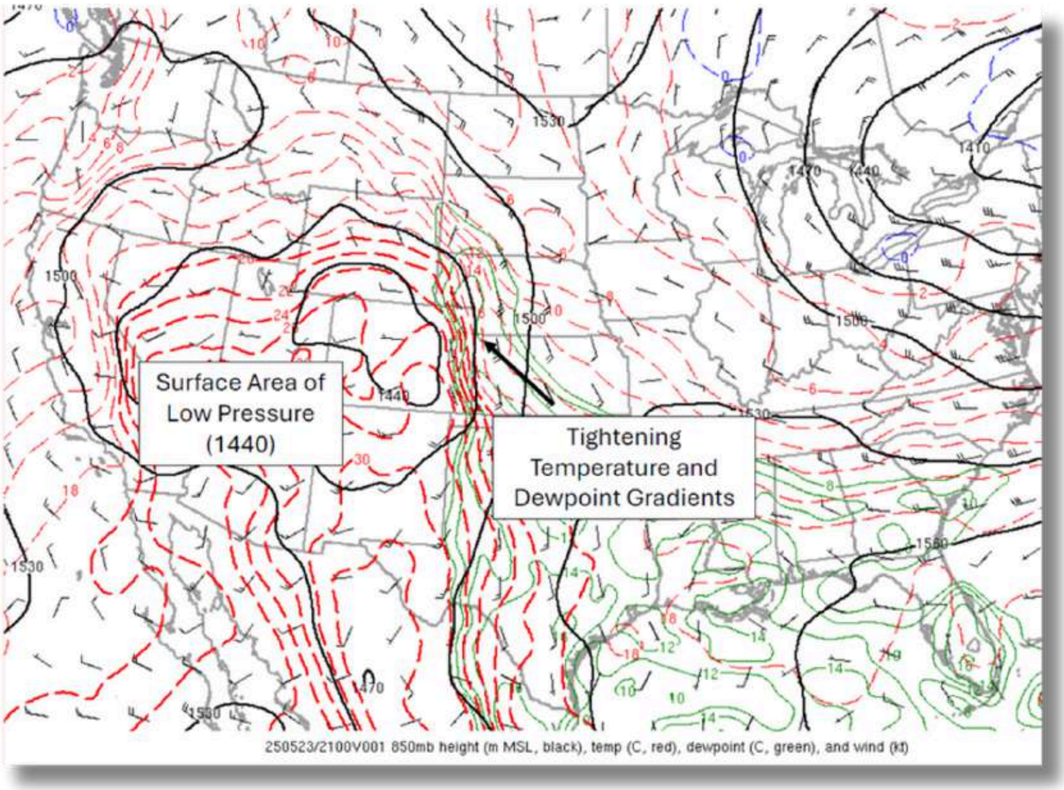

Figure 22: 850mb Height, Temperature, and Dewpoint (2100z)

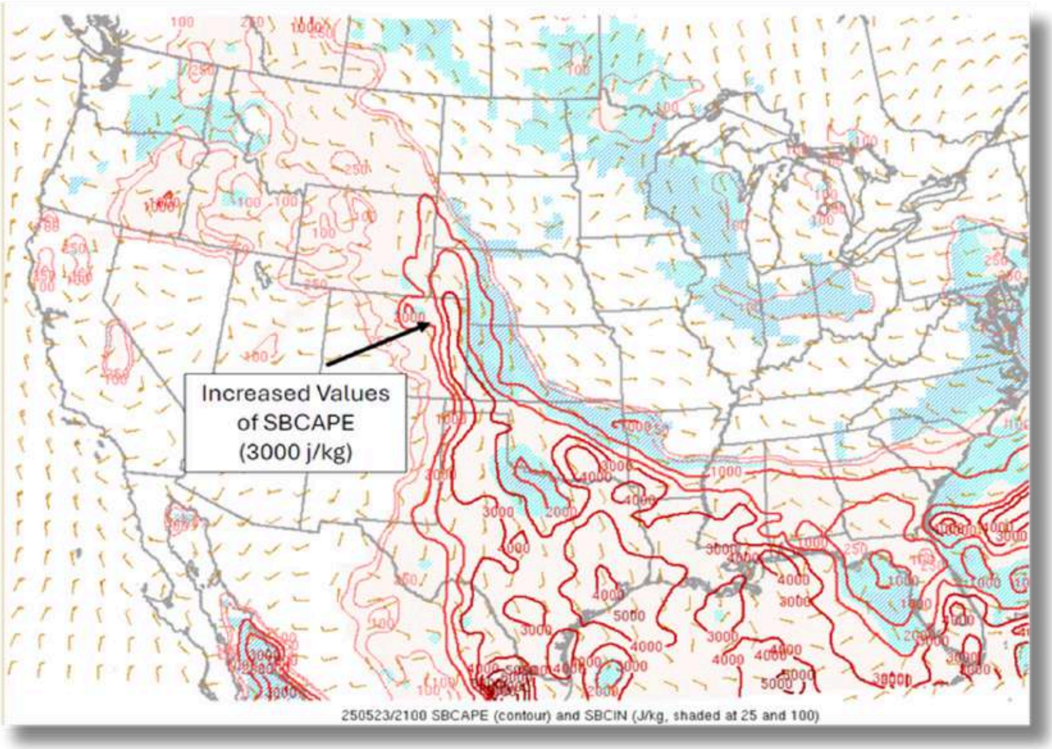

Figure 23: Surface Based CAPE and CIN (2100z)

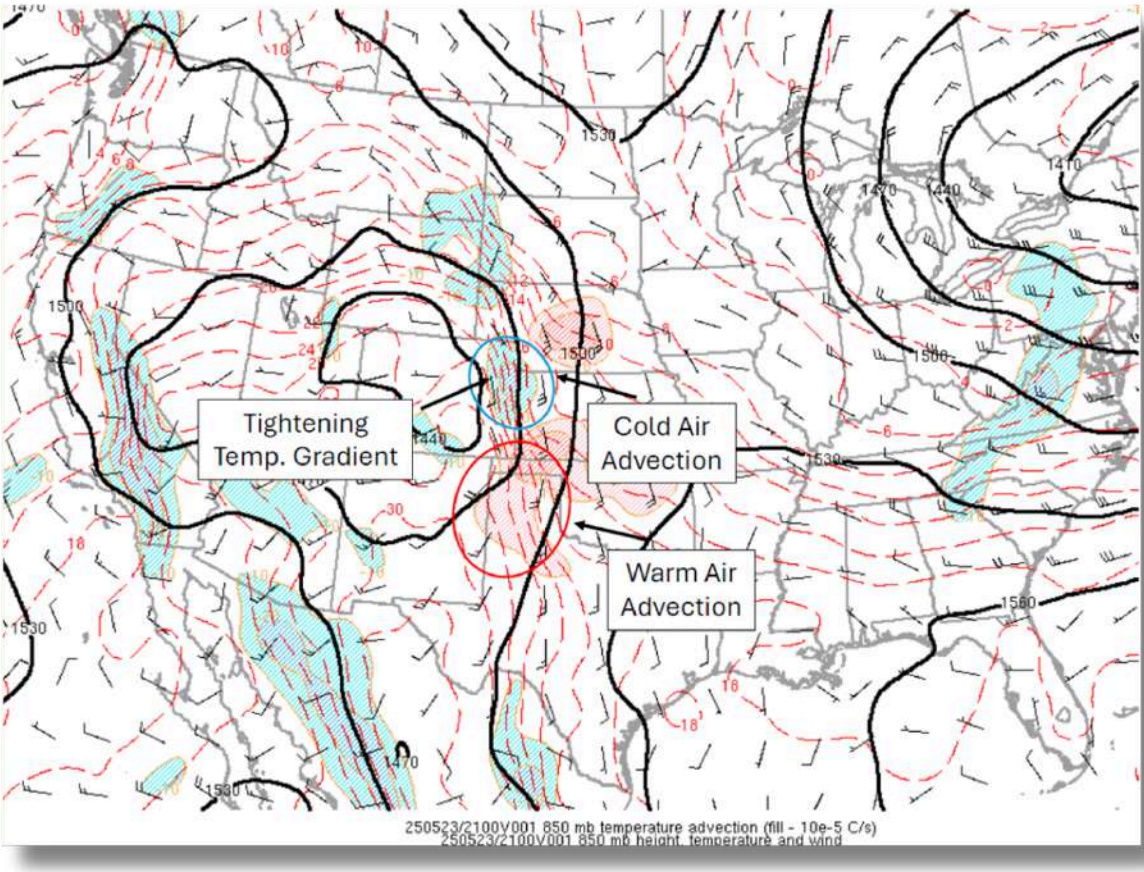

Figure 24: 850 mb Temperature Advection (2100z)

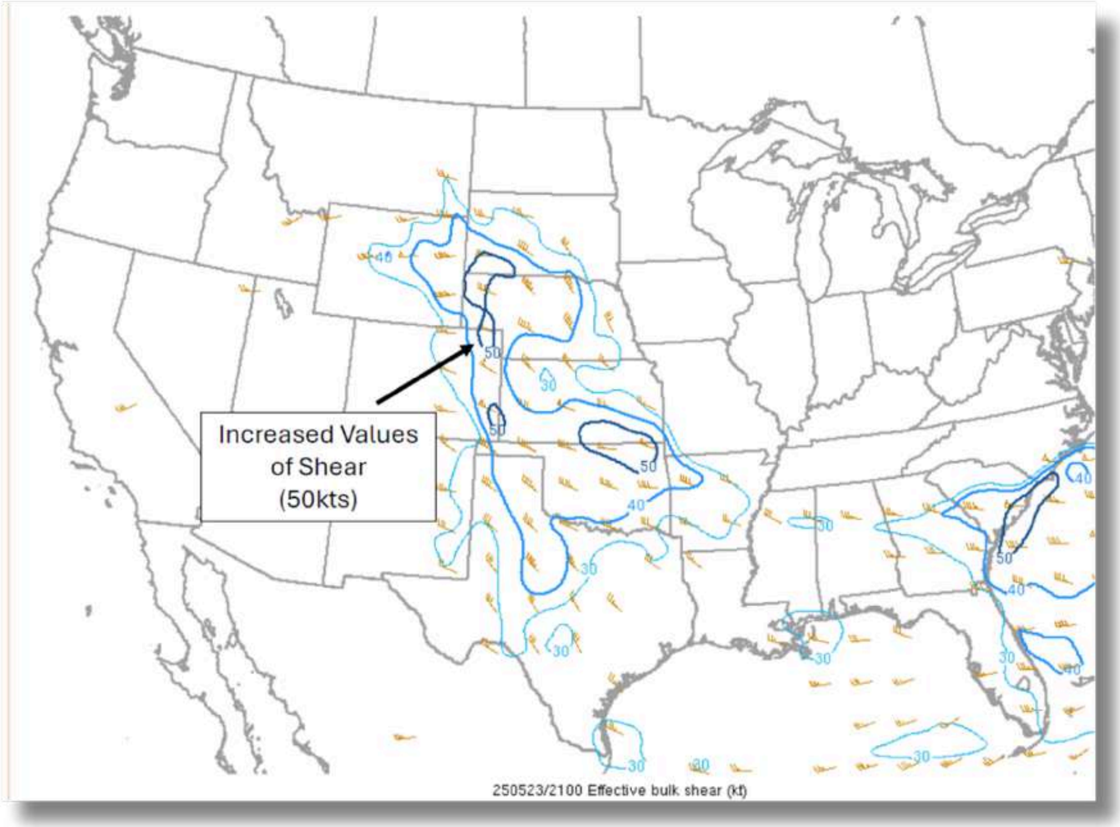

Figure 25: Effective Bulk Shear (2100z)

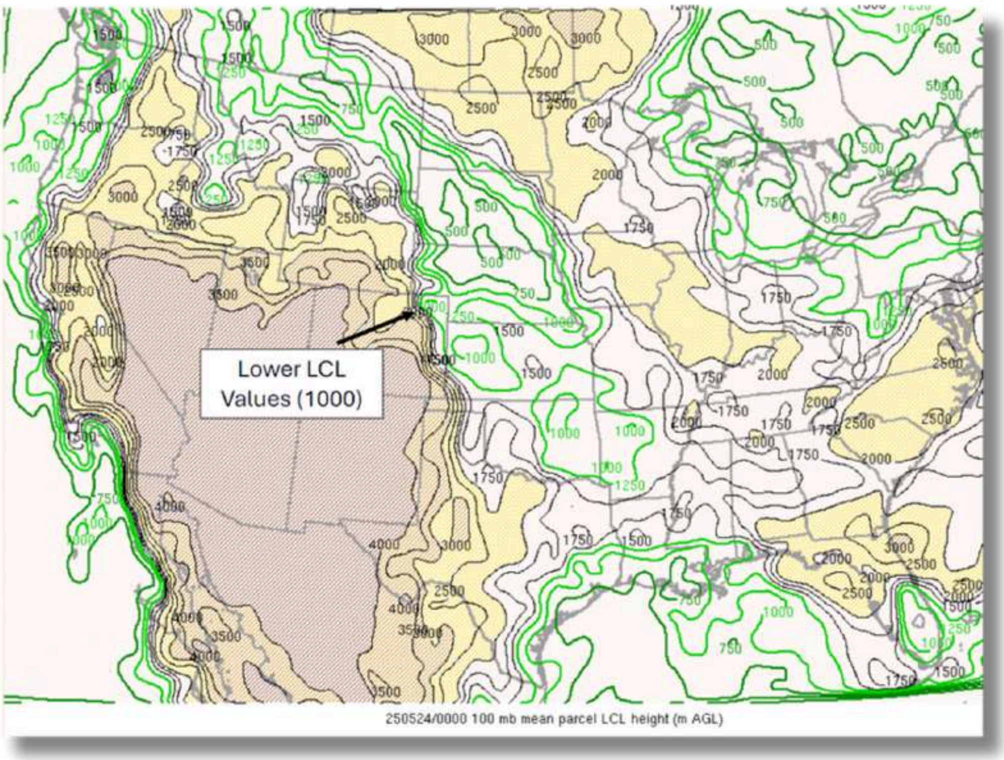

Figure 26: 100mb Mean Parcel LCL Height (0000z)

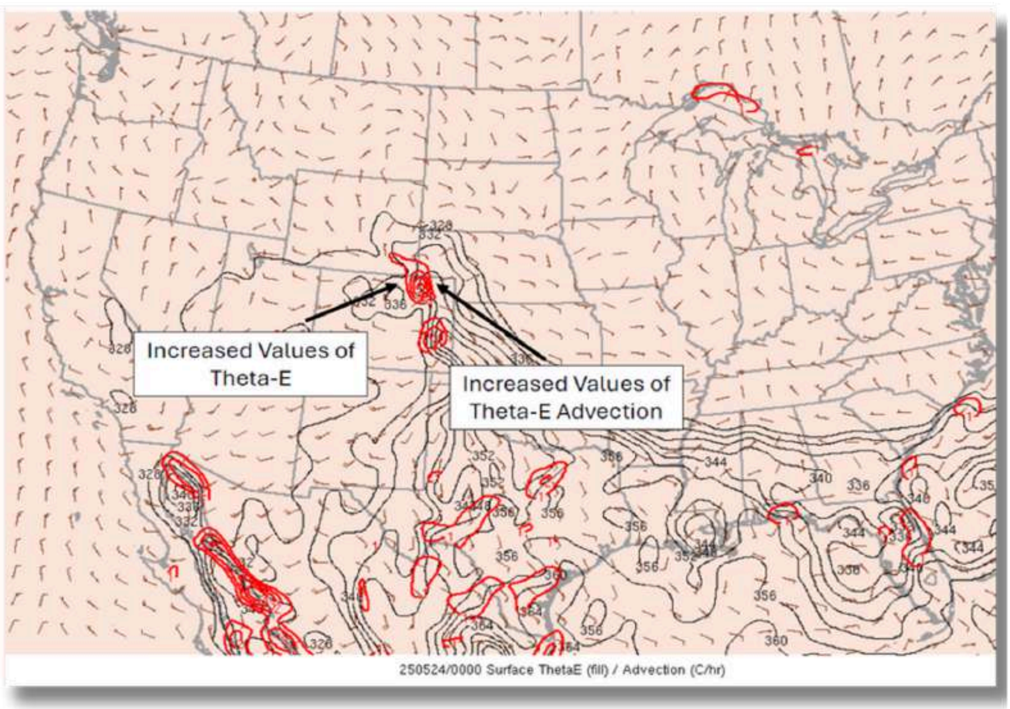

Figure 27: Surface Theta E and Advection (0000z)

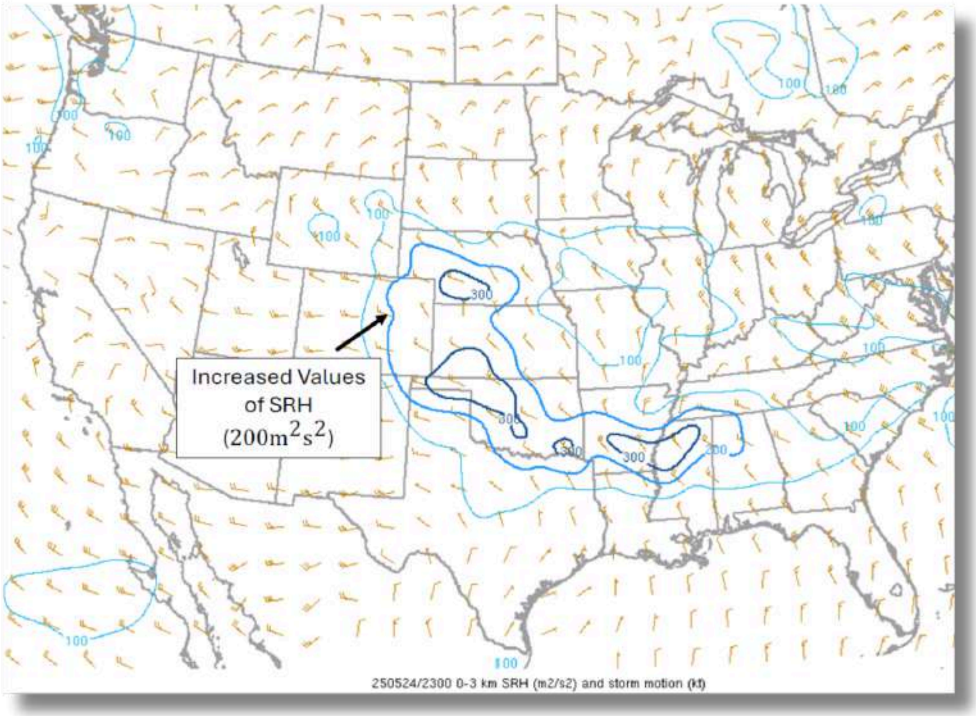

Figure 28: 0-3km Storm Relative Helicity

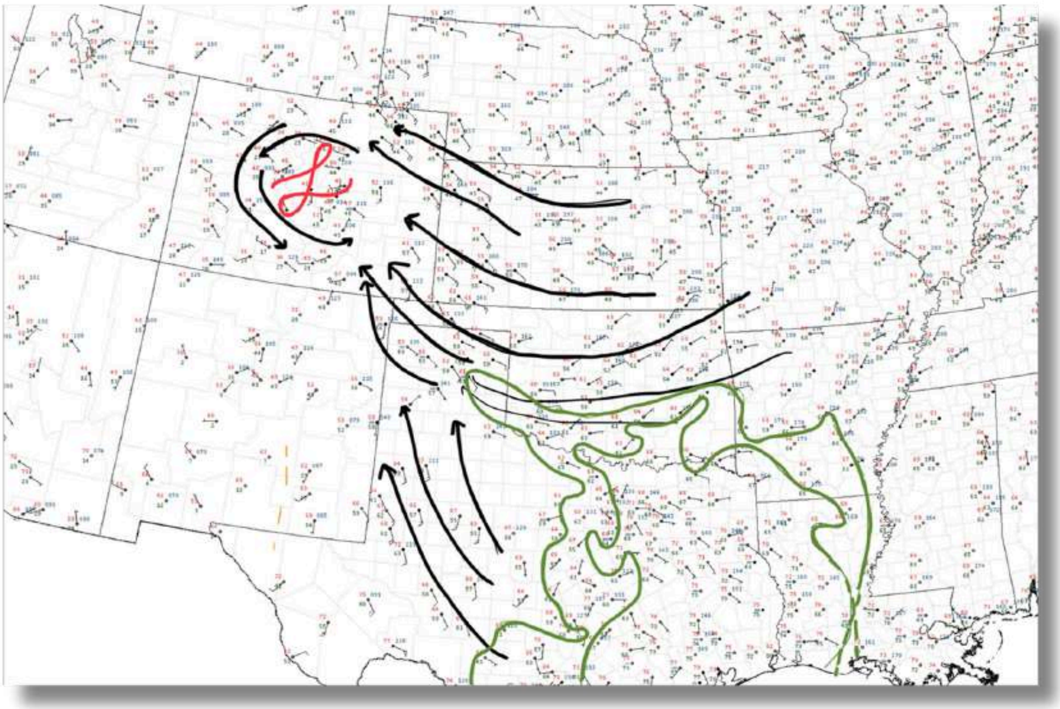

Figure 29: Hand surface analysis from May 23, 2025 at 12z.

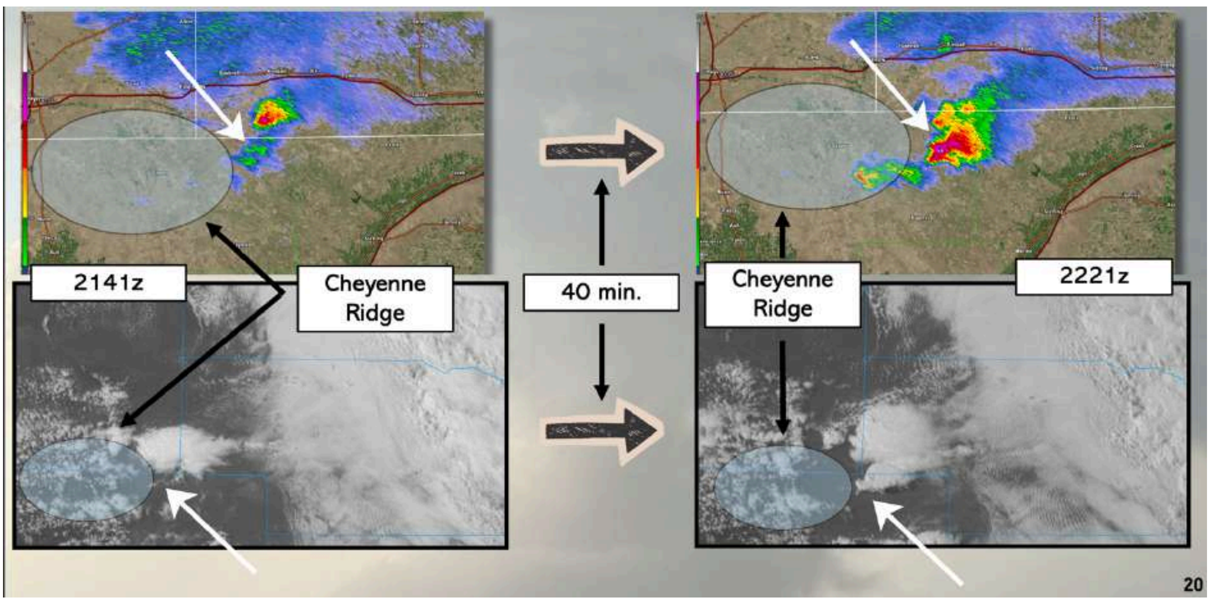

Figure 30: Satellite and Radar imagery showing the influence of the Cheyenne Ridge

Figure 31: Upper-Level Jet Streak dynamics graphic

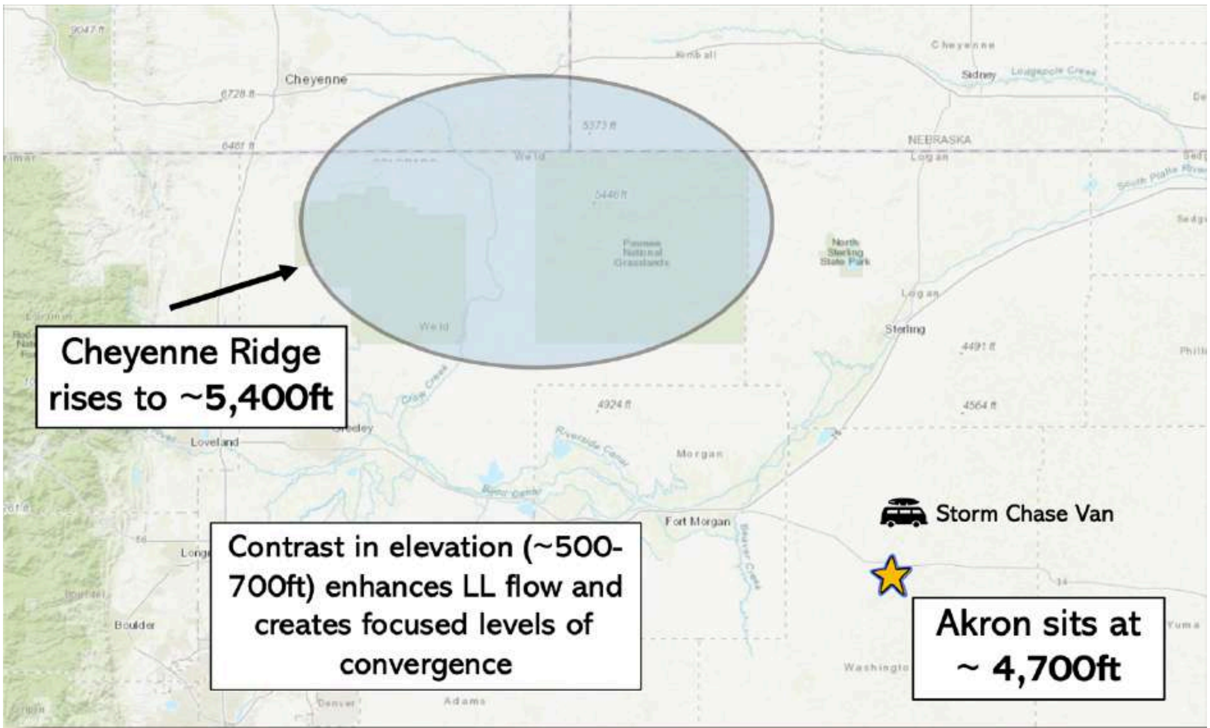

Figure 33: Location reference map

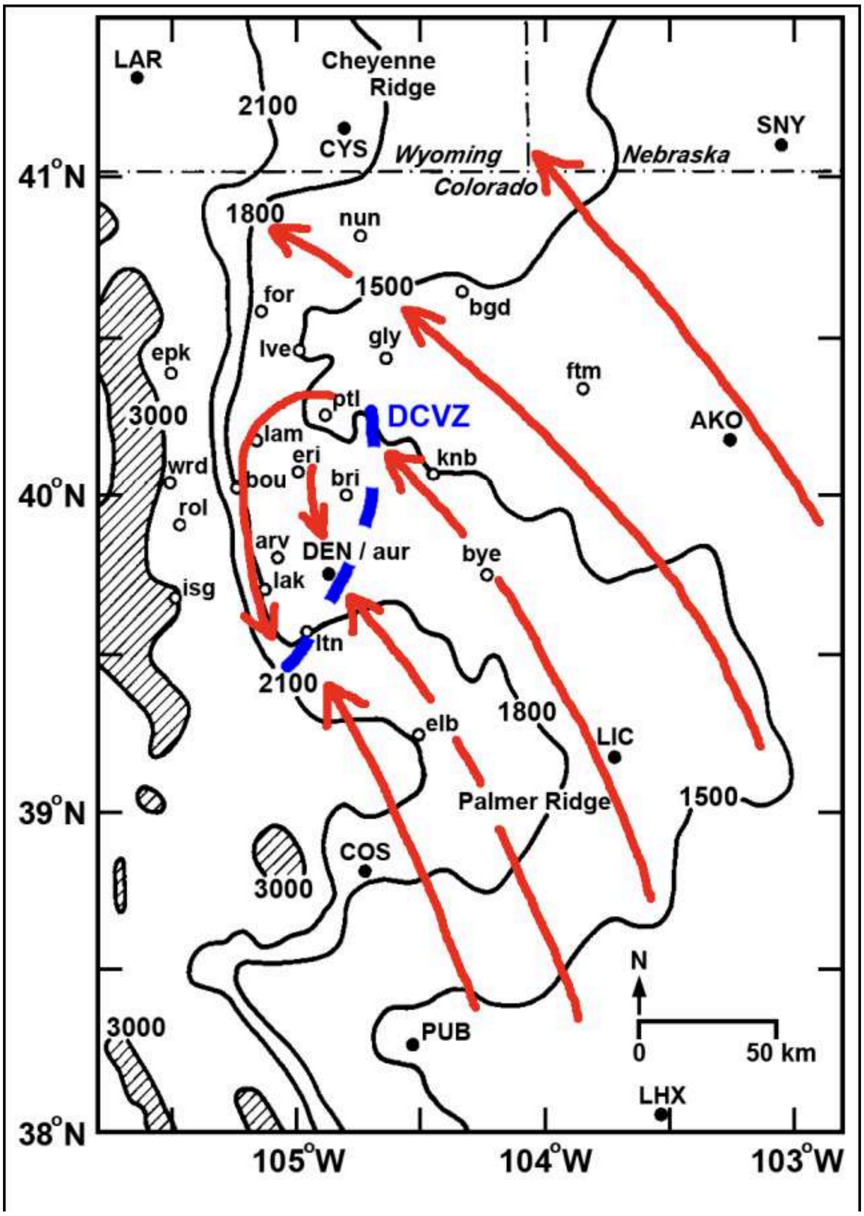

Figure 34: The Denver Convergence Vorticity Zone

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